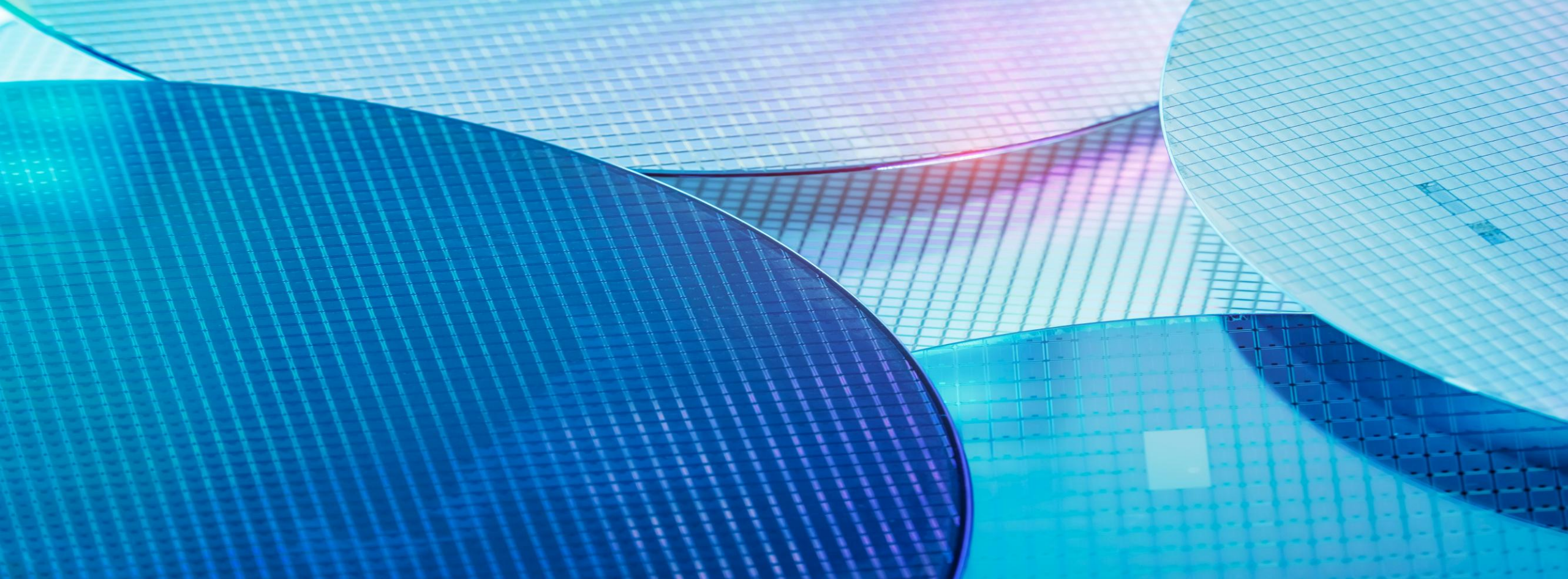
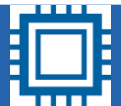




E2E Semiconductor Foundry Workflow Training

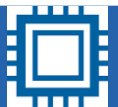
v1.0 (14-Mar-2025)

Charu Maurya + Aniket Agshikar



Disclaimers

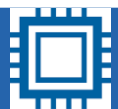
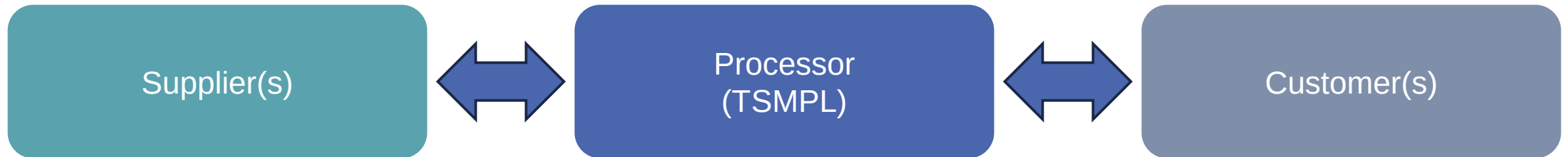
- This high-level generic training document is developed based on the best of our industry knowledge
- This training deck is intended for general use only and is not meant for any monetary purpose
- We do not undertake any responsibility for errors or omissions in the content
- Users are advised to verify the information before applying it in real-world scenarios
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3 parts of any business organization

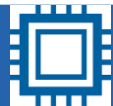
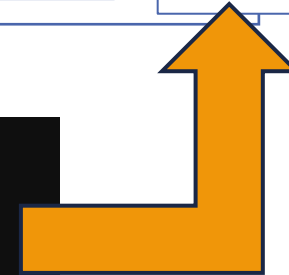
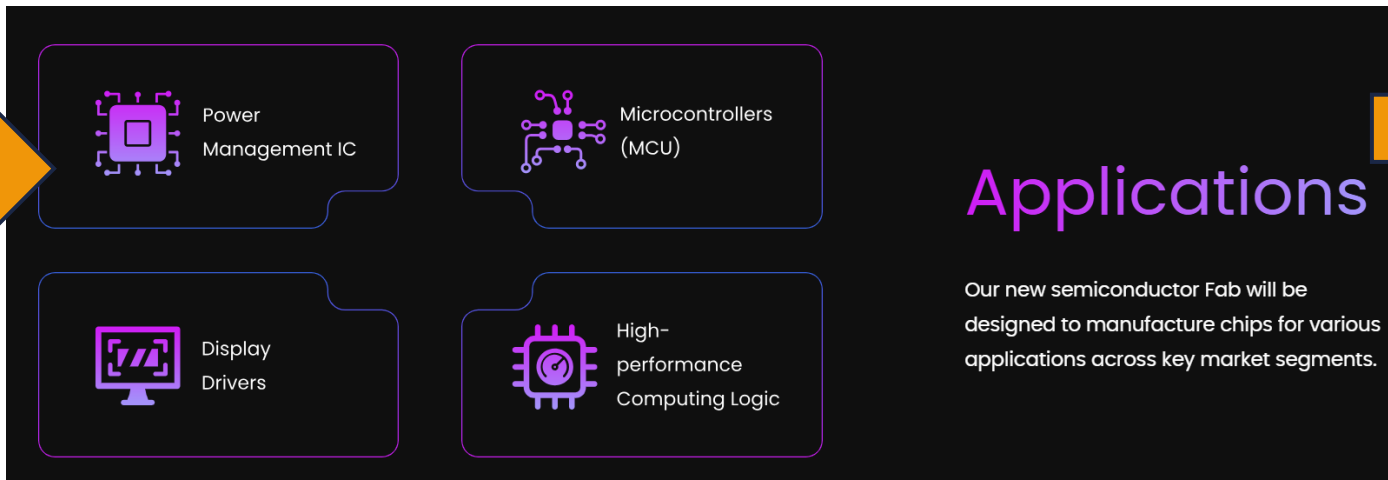
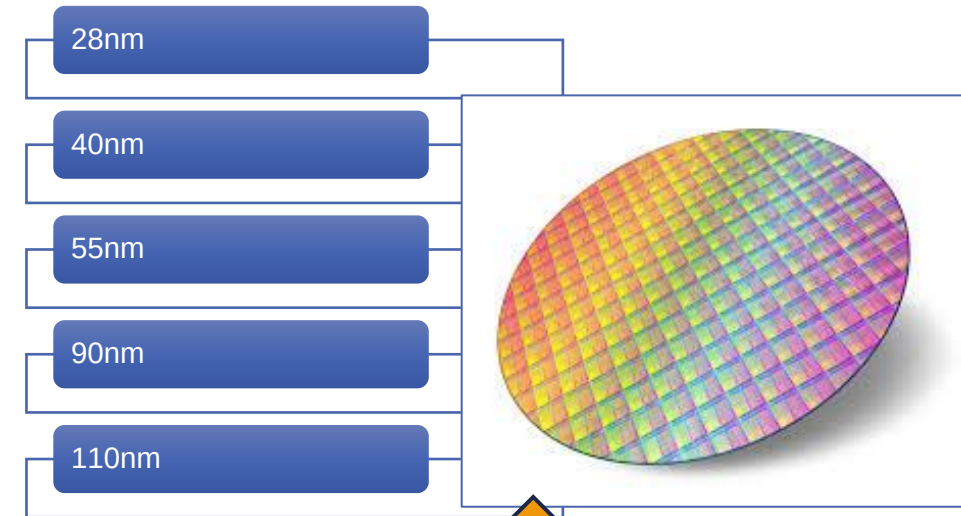
Every business needs to understand their customers' needs and provide solutions which meet their requirements.

To make the products/services for our customers, we will need suppliers who provide us raw materials, hardware, software, services, etc.



Our products are semiconductor process technologies

At TSMPL, we will be developing advanced semiconductor process technologies (e.g. 28/40/55/90/110nm) to cater to various market segments and application space



Types of semiconductor technology offerings

1st Source is a platform technology which has all the features & open to all customers. 2nd Source is a copy-cat technology which will have limited Design Enablement & IP ecosystem. Custom Technology is a co-developed with the customer

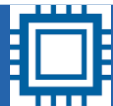
1st Source



2nd Source: Copycat

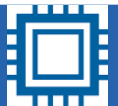
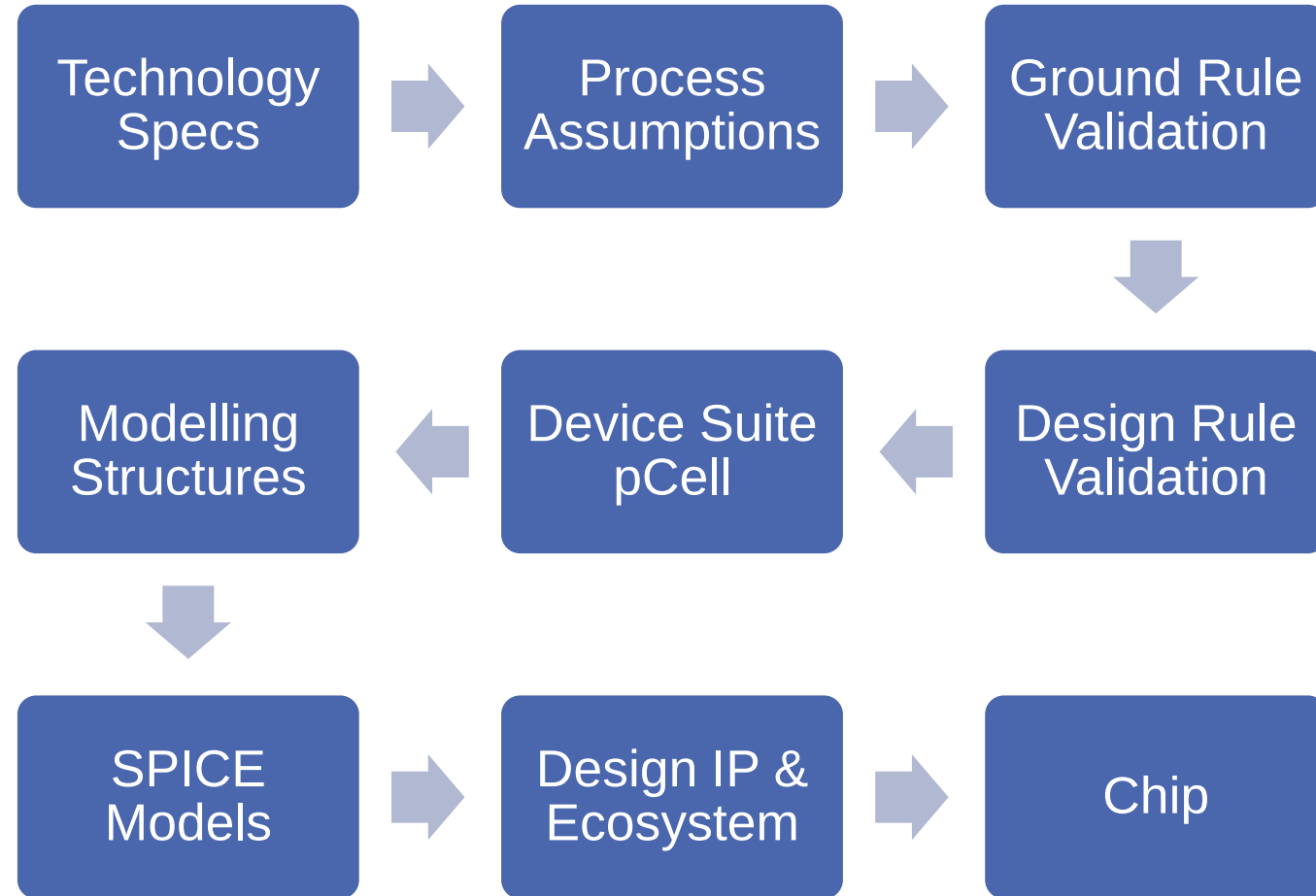


Custom Technology

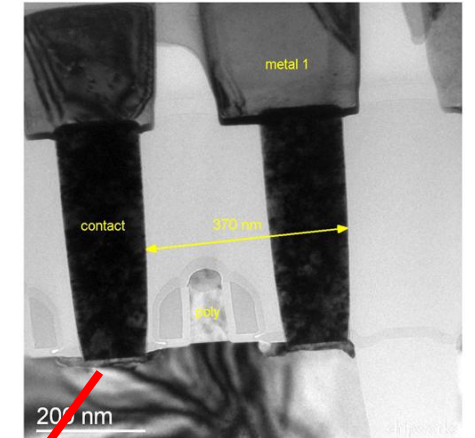
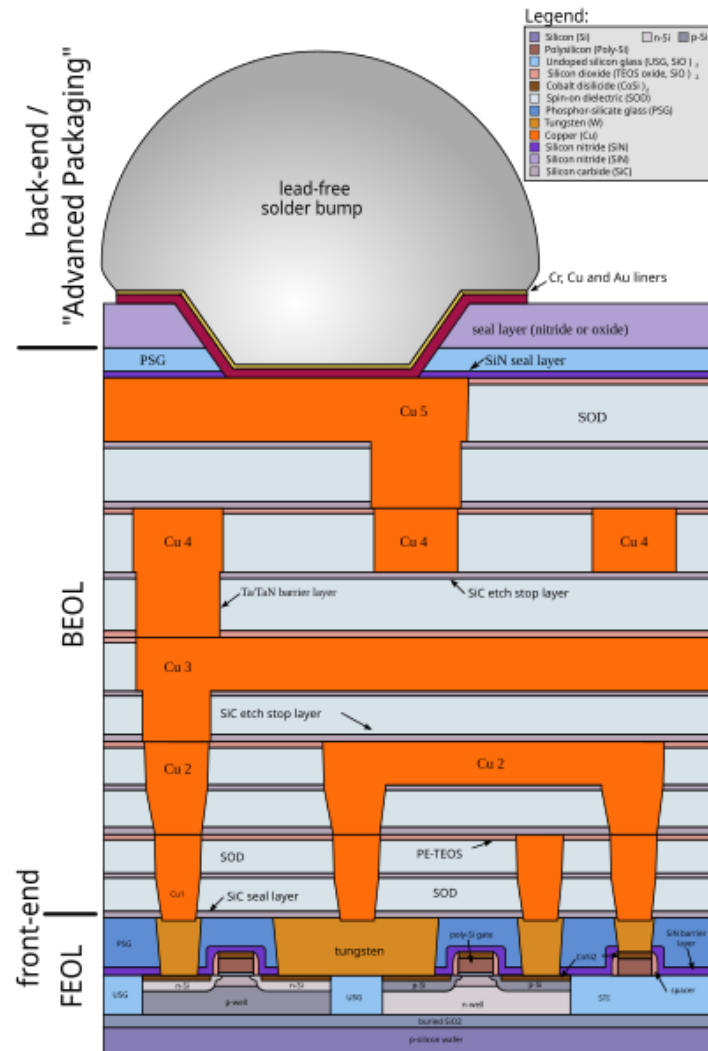


1st Source Technology Development Flow

Starting with the Technology Specifications to final chip: each step is critical to build a robust technology offering!



Semiconductor wafer process groups: FEOL, BEOL & Far-BEOL

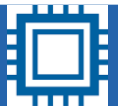
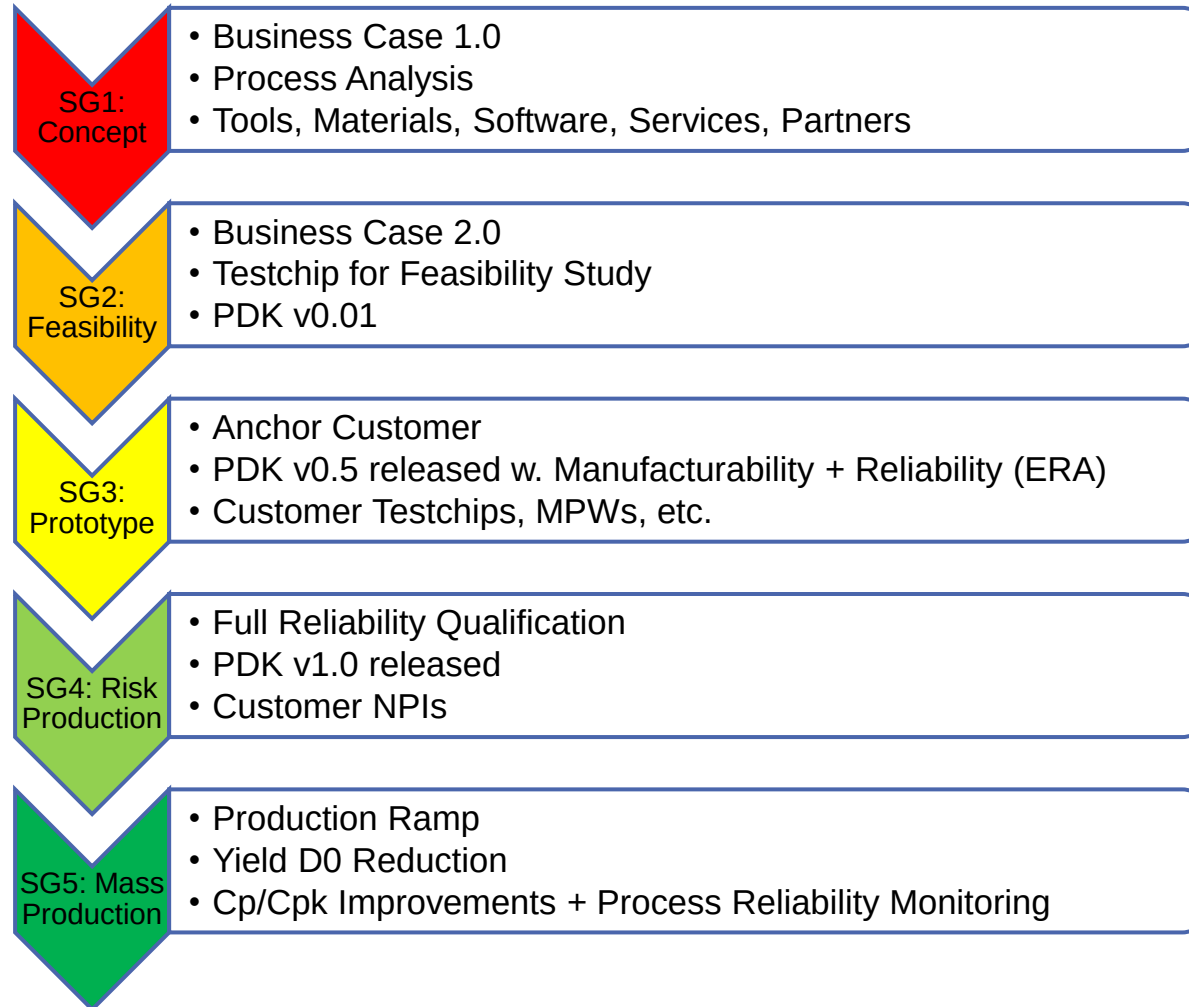


Sources: [A Review of TSMC 28 nm Process Technology | TechInsights](#) ; Wikipedia, Google Images (Intel)

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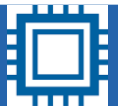
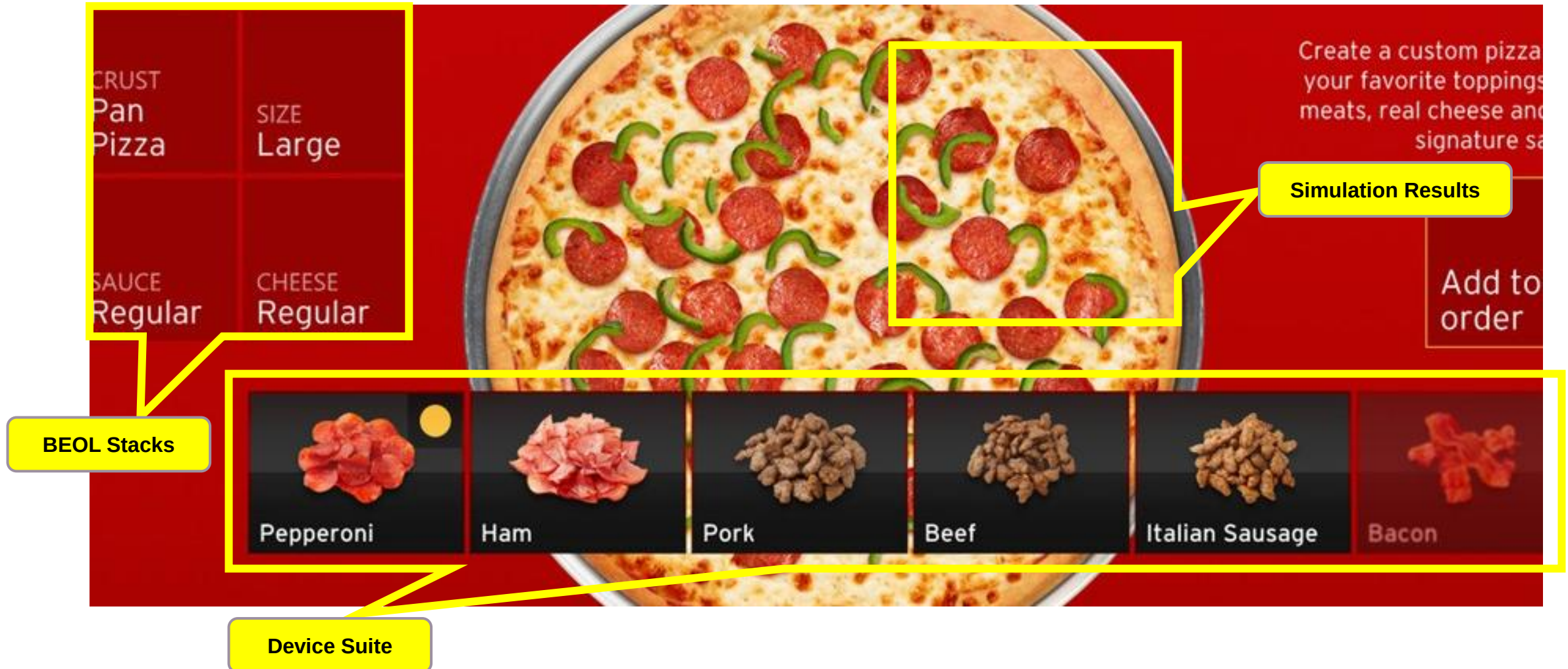
Technology Qualification Process

Semiconductor process technology gets qualified in stage-gate phases as per industry standards requirements (e.g. ISO, IATF16949, JEDEC, AEC, AIAG). As the process gets matured, yield and Cp/Cpk improves!



What is a PDK? How is it connected with the Fab?

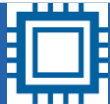
PDK (Process Development Kit) is the software which simulates the foundry's process to produce the wafer (pizza). It enables the designers to simulate and check how the chip will work within the boundary conditions set by the Fab through the PDK (e.g. DRs, PEX, etc.)



What is a Design Manual (DM)?

Design Manual is a document within the PDK which specifies what are the various key features of the process technology, the mask layers, the design rules, the devices and BEOL stacks offered, Mask Booleans, etc.

- The devices and BEOL stacks are the various building blocks to build chips
- Just like on a Lego box, there are a set of fixed blocks and rules. But you can use your creativity to build a ship or a house or even a spaceship!
- Similarly, same semiconductor process technology could be used for designing various applications like processors, communication chips, ASICs, etc.



Design Workflow

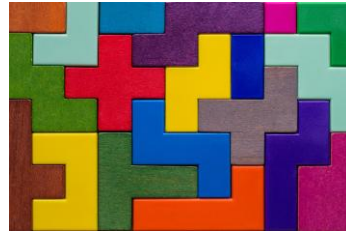
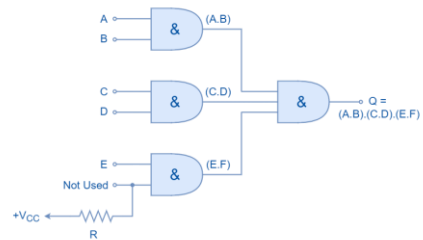


Requirement & Specification

- Product Definition
- Planning & Budgeting
- Design Integration
- Industry Compliance

- RTL Design
- RTL Verification

RTL Design & Verification

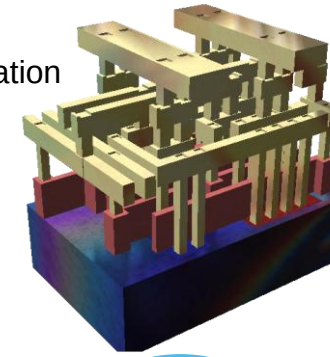


Synthesis & Place & Route (P&R)

- Logic Synthesis
- Gate Level Netlist
- Physical Design

- DRC
- LVS
- PEX
- Post-layout Circuit Simulation

Physical Verification

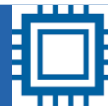
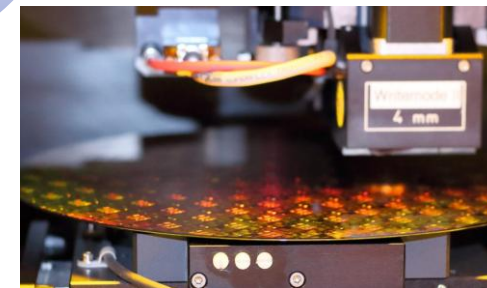


Tapeout

- GDSII Generation
- Mask Generation
- OPC Checks

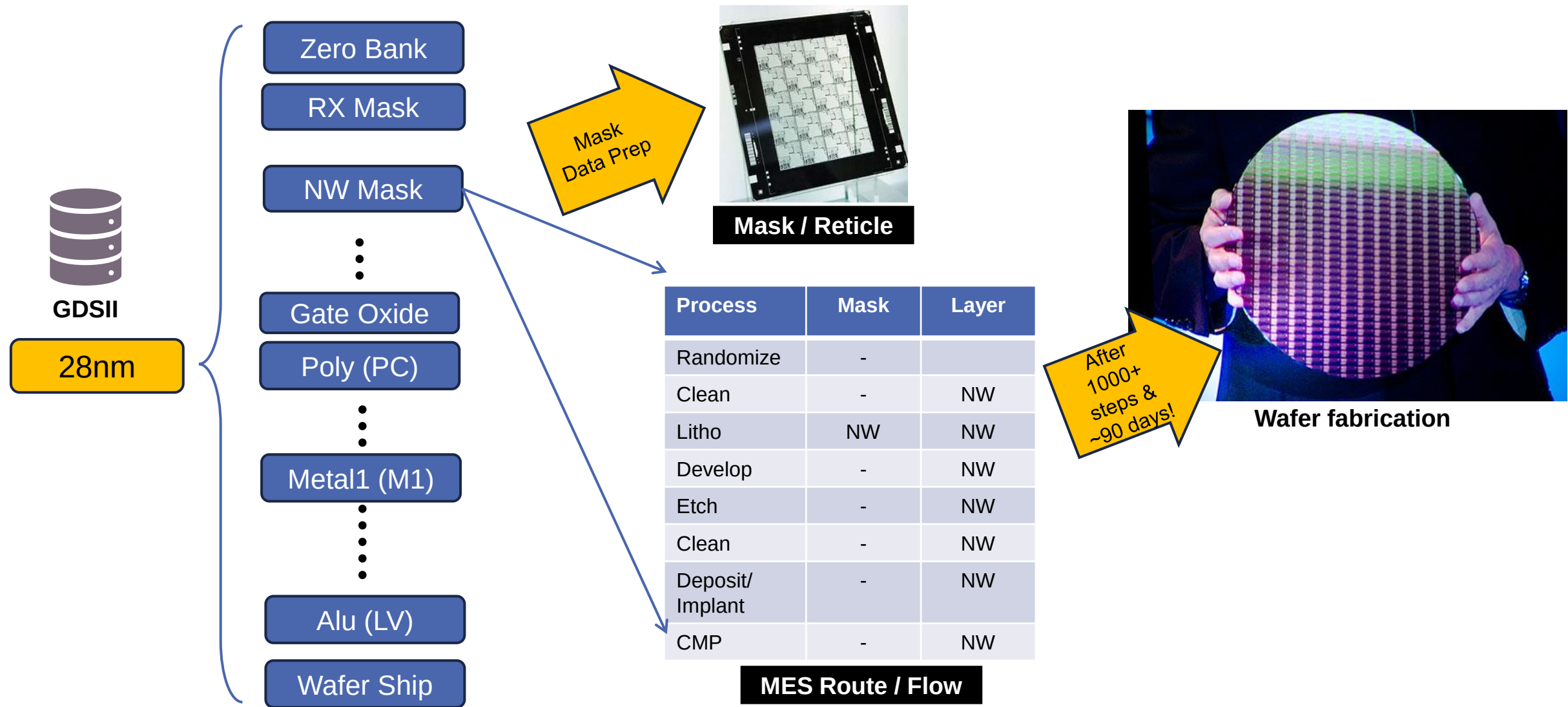
- Wafer fabrication
- Electrical & Logic testing

Fabrication & Testing



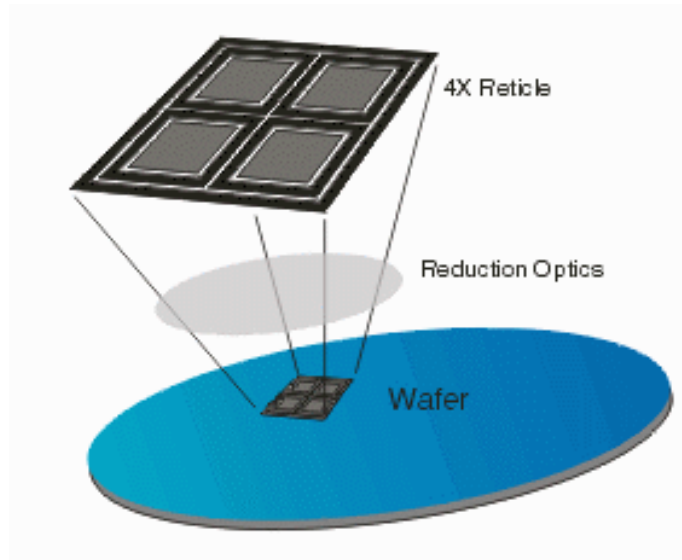
Tapeout process & Mask Layers

During the Tapeout process, the input GDS file gets decomposed into individual lithography mask layers based on the specific technology and Booleans. The number and type of mask is dependent upon the technology & metal levels.

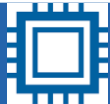
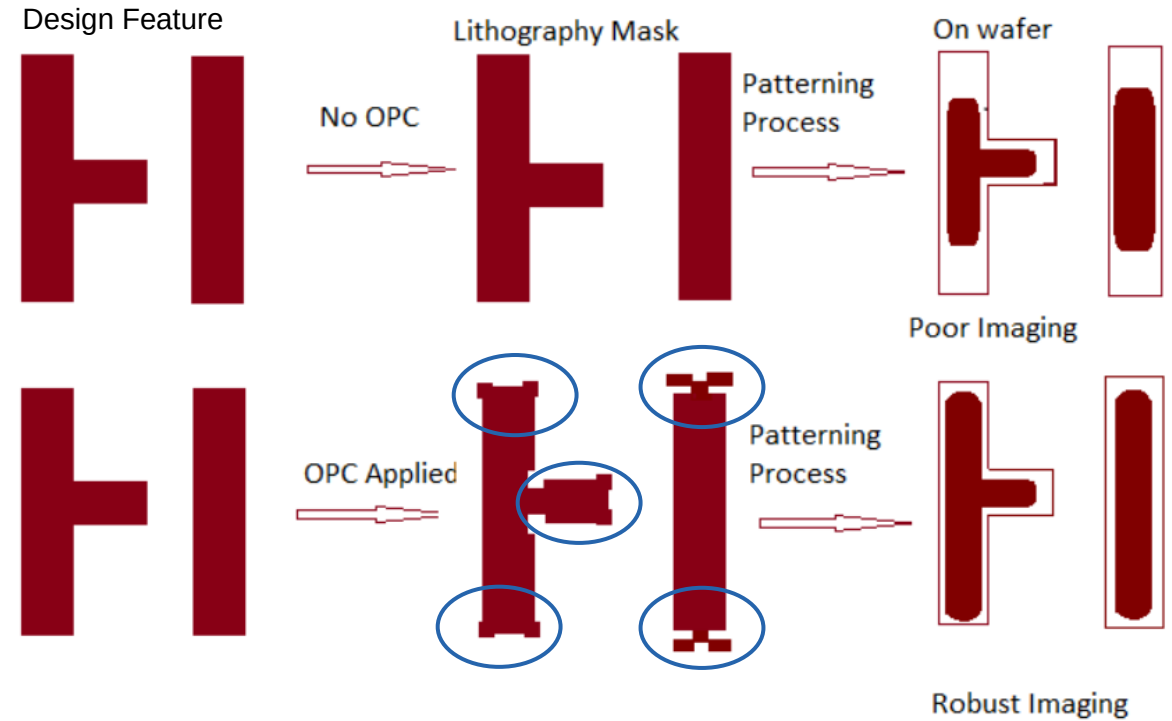


Mask writing and OPC (Optical Proximity Correction)

Printing the sub-micron features on silicon wafer has certain optical limitations which can be overcome by OPC. It adds some extra shapes at the sharp corners and edges to improve lithographic process



Writing mask on wafer

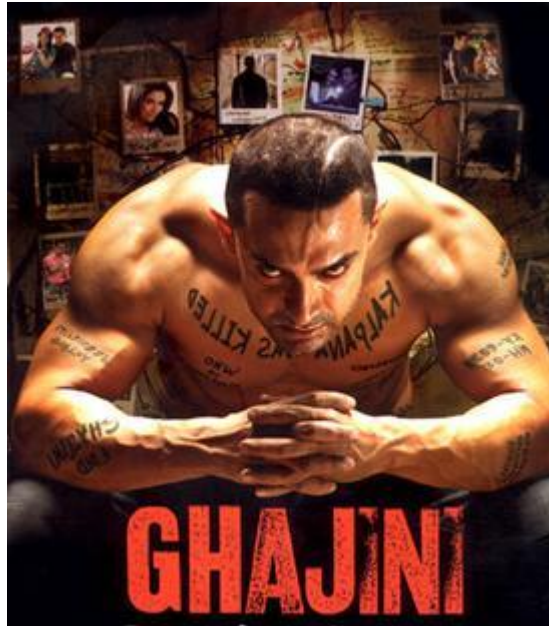
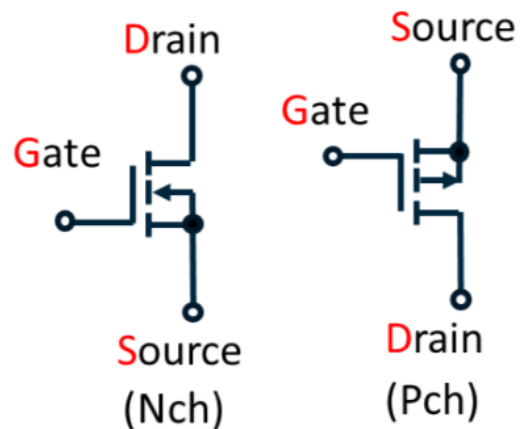
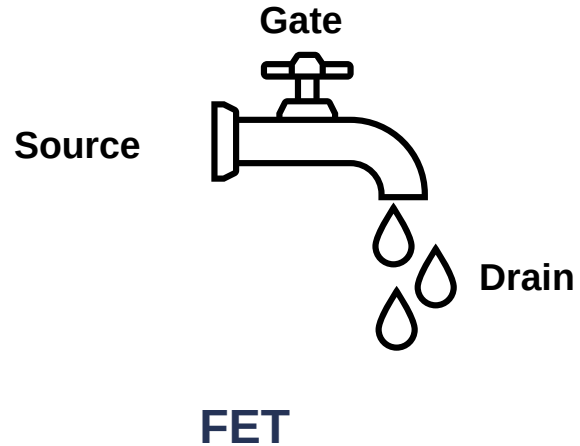


Devices: FETs, SRAM, NVM

FET: Field Effective Transistor is widely used in electronic devices for switching and amplifying signals.

SRAM (Static RAM) retains data bits in its memory if power is being supplied.

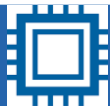
Non-Volatile Memory (NVM) can retain stored information even after power is removed.



What do you think?

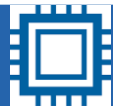
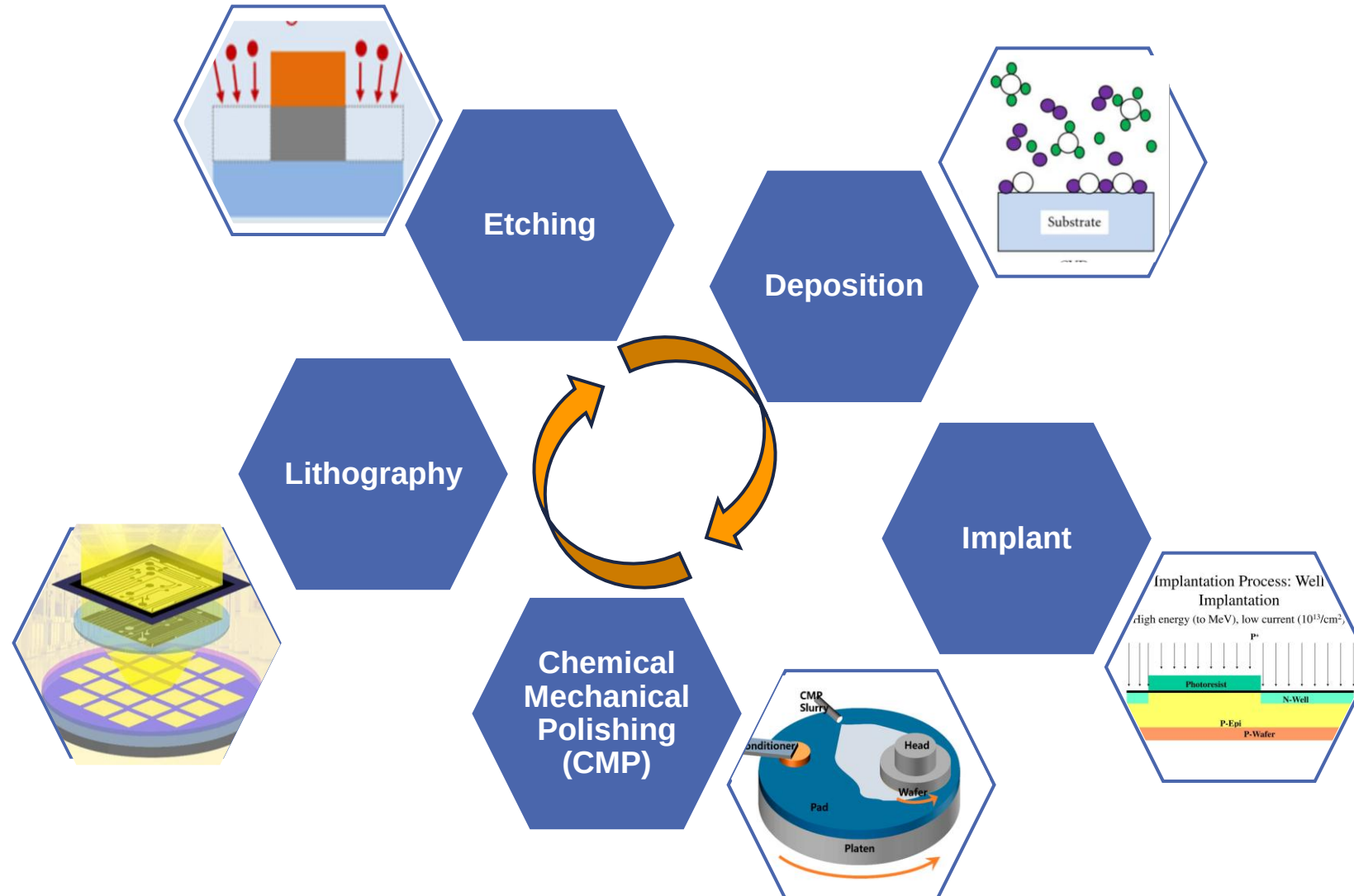


SIM card: NVM or SRAM?



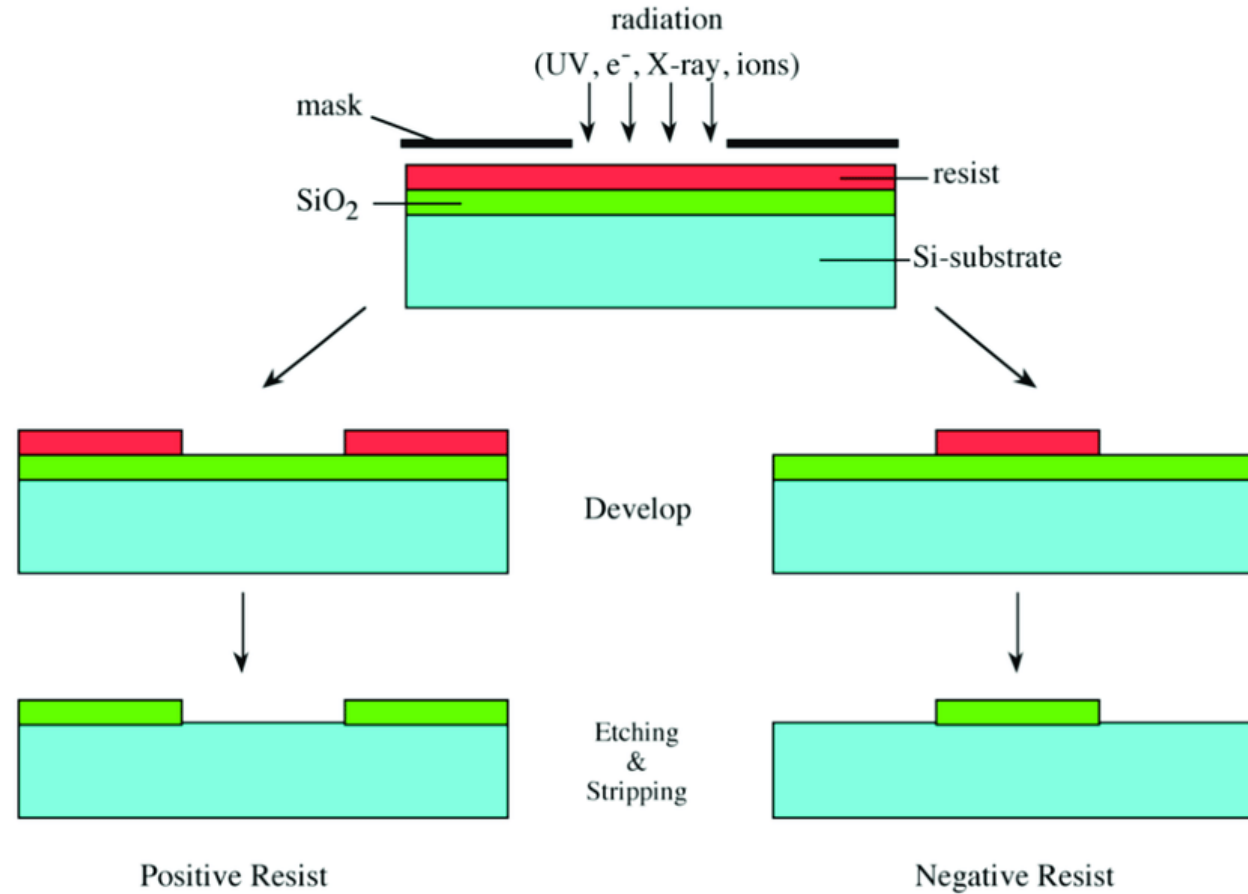
5 Critical Semiconductor Process steps

Semiconductor manufacturing involves mostly the following 5 steps, which are repeated through various process layers to build various devices, BEOL stacks, etc. which eventually delivers a fully processed wafer

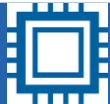


Photolithography / Lithography

Generates designed layer patterns by passing special light via lenses through photomasks onto wafers coated with light-sensitive photoresist

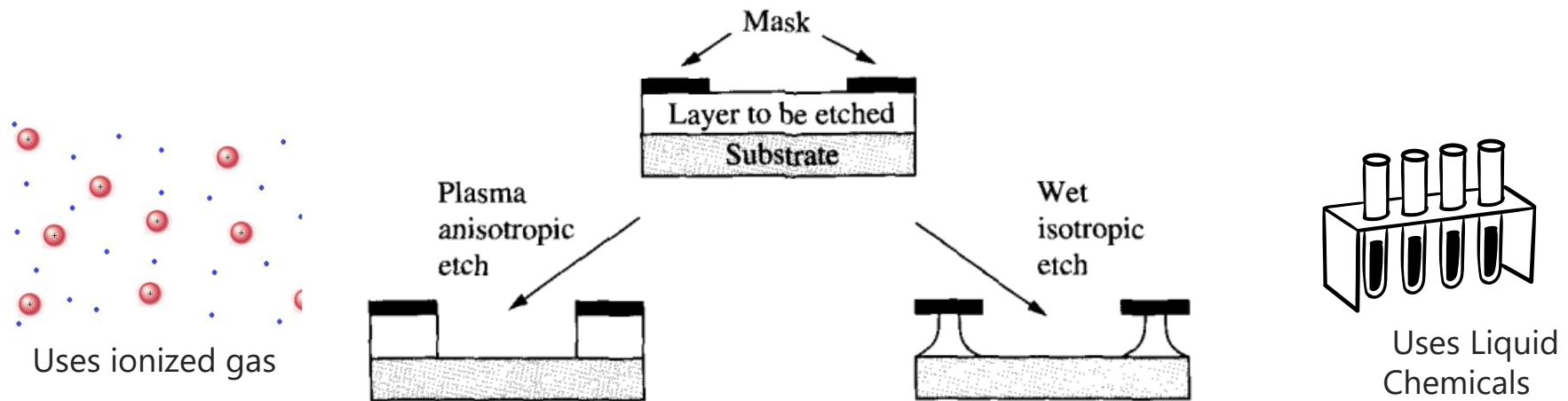
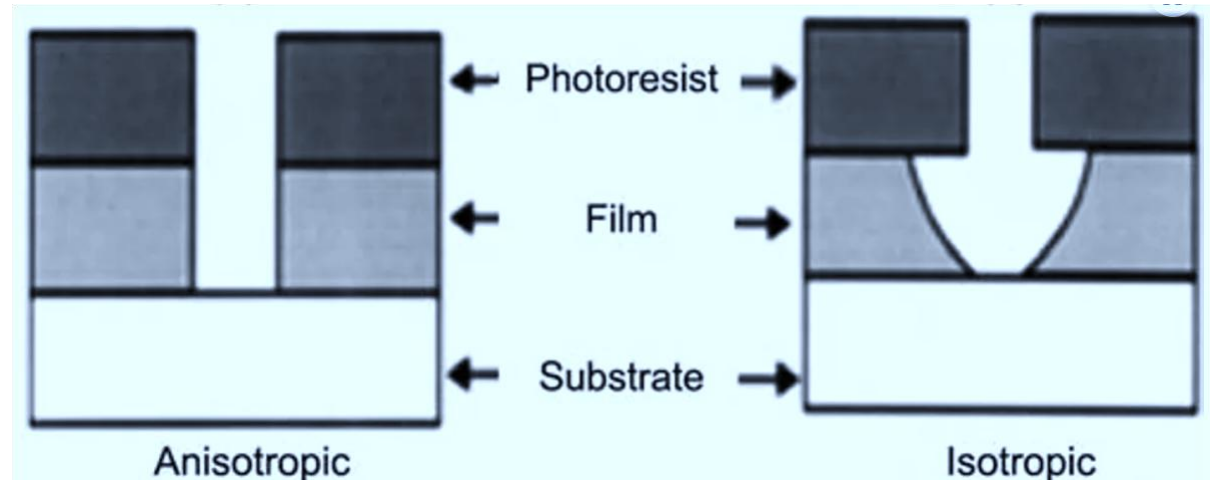


Sources: [Comparison of positive and negative photoresists](#) By OpenStax (Page 3/3) | Jobilize



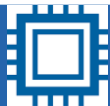
Etching

Etching selectively removes unwanted layers by use of wet, dry or plasma chemistry



Sources: [Dry Etching vs Wet Etching: A Comprehensive Comparison](#)

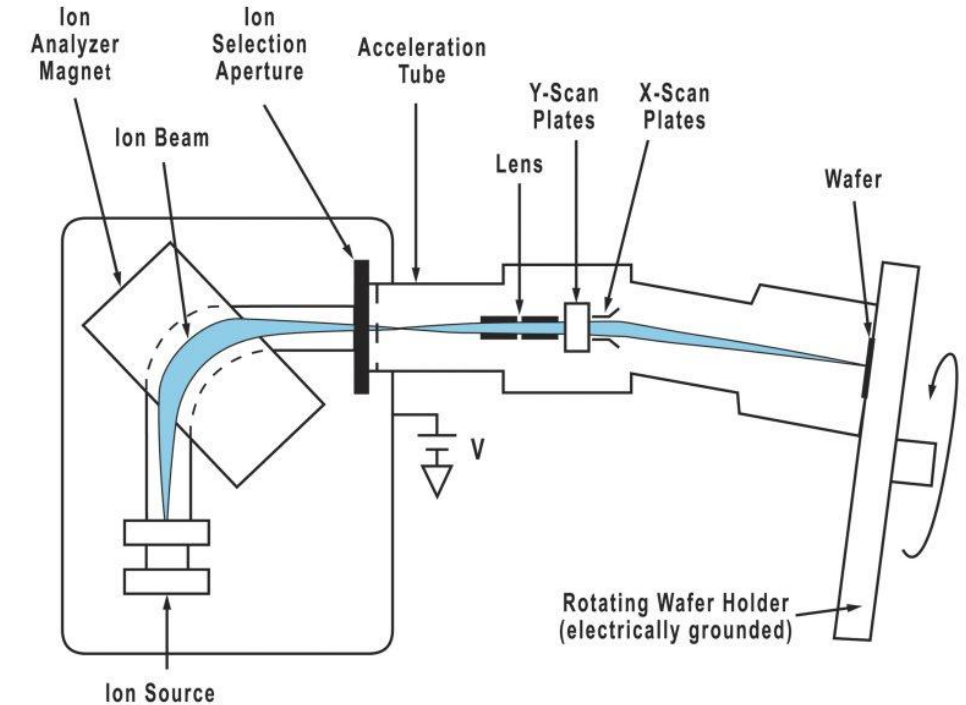
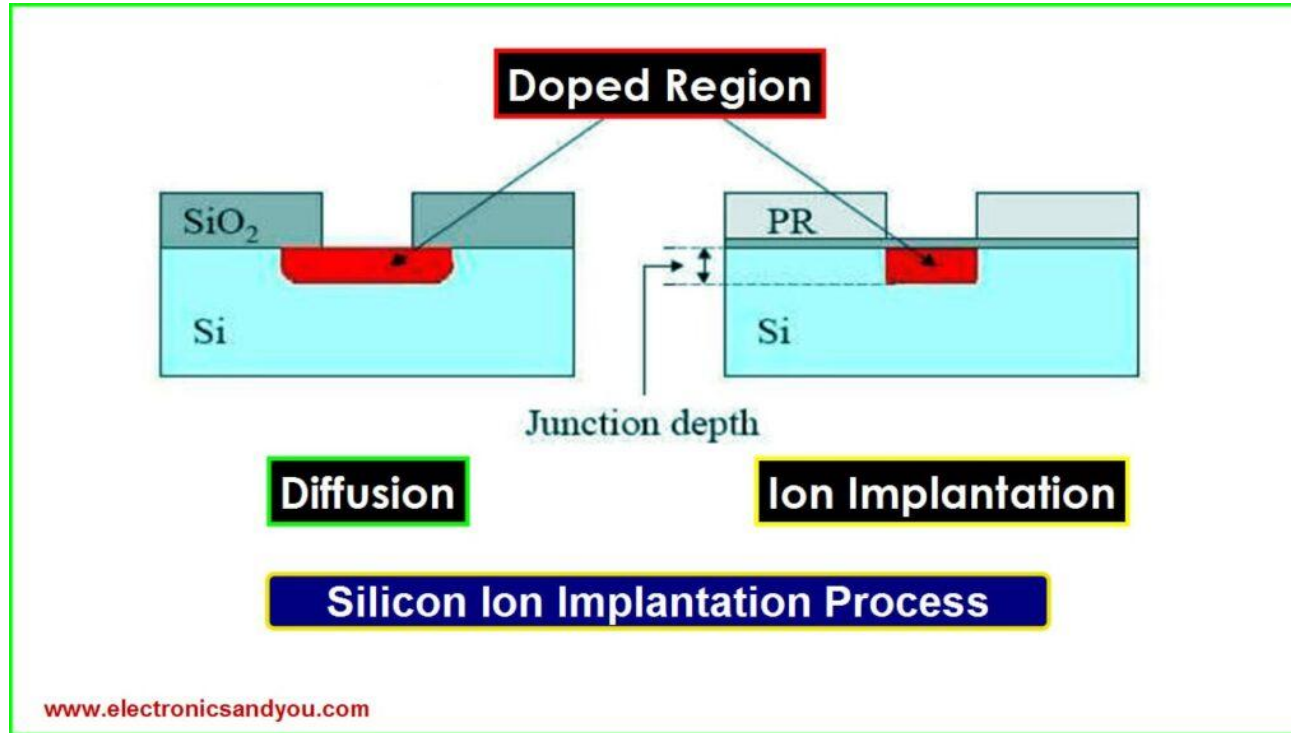
https://www.researchgate.net/figure/Transfer-of-a-pattern-to-a-photosensitive-material-and-selective-etching_fig97_299345334



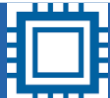
Implant (MC/HC, High Energy)

Ion implantation injects n/p type dopant species into the silicon to alter its conductive properties

- High-current(HC) implanters are used for source/drain development
- Medium-current (MC) tools are used for well implants
- High-energy (HE) implanters are used for deep well implants

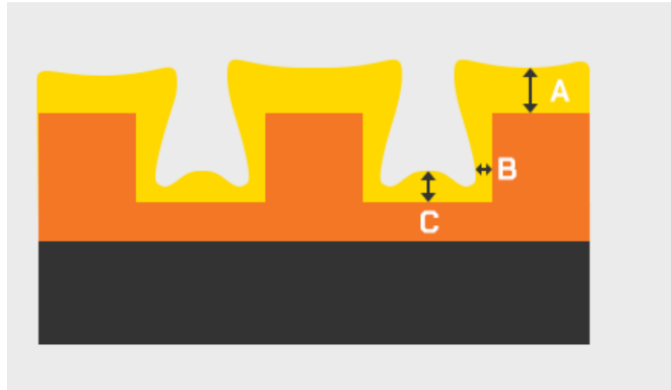


Source: <https://www.electronicssandyou.com/blog/semiconductor-manufacturing-process-steps-and-technology-used.html>



Deposition

Deposition is a process that deposits a thin layer of material on a surface of the wafer usually via Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD), Electro-Chemical Deposition (ECD), Atomic Layer Deposition (ALD), etc.



A,B,C: Height & width of a material being deposited – Conformality issue for lower nodes!

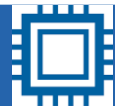


*Different flavoured layers in cake,
Different material layer in a wafer!*

ALD Process (SiO_2 Example)



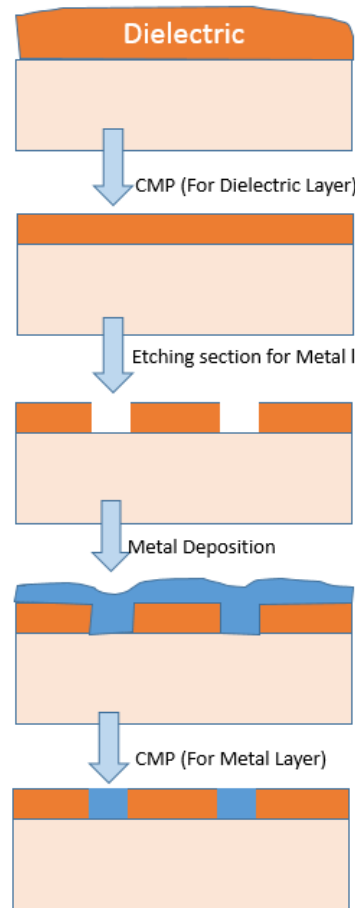
Source: <https://semiengineering.com/a-look-at-atomic-layer-deposition-2/>



CMP (Chemical Mechanical Planarization/Polishing)

Chemical and/or mechanical polishing to achieve wafer surface planarization to ensure flat topology for the next process step

Planarization (flattening) of the wafer topology for the next Litho mask step or process

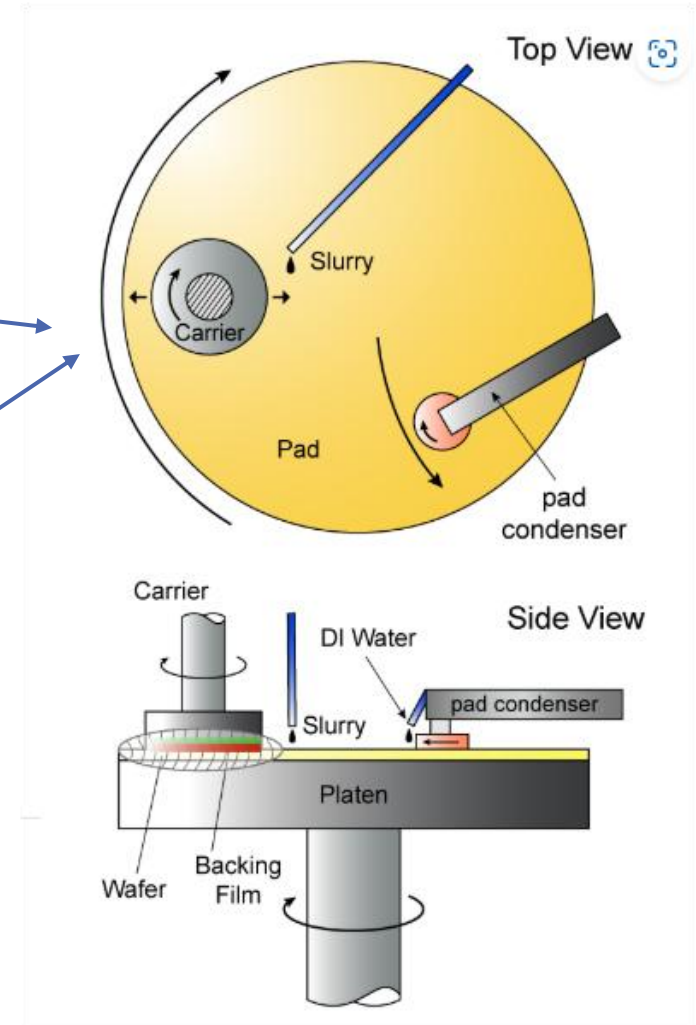


Step1

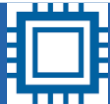
Step2

Step3

Step4

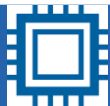
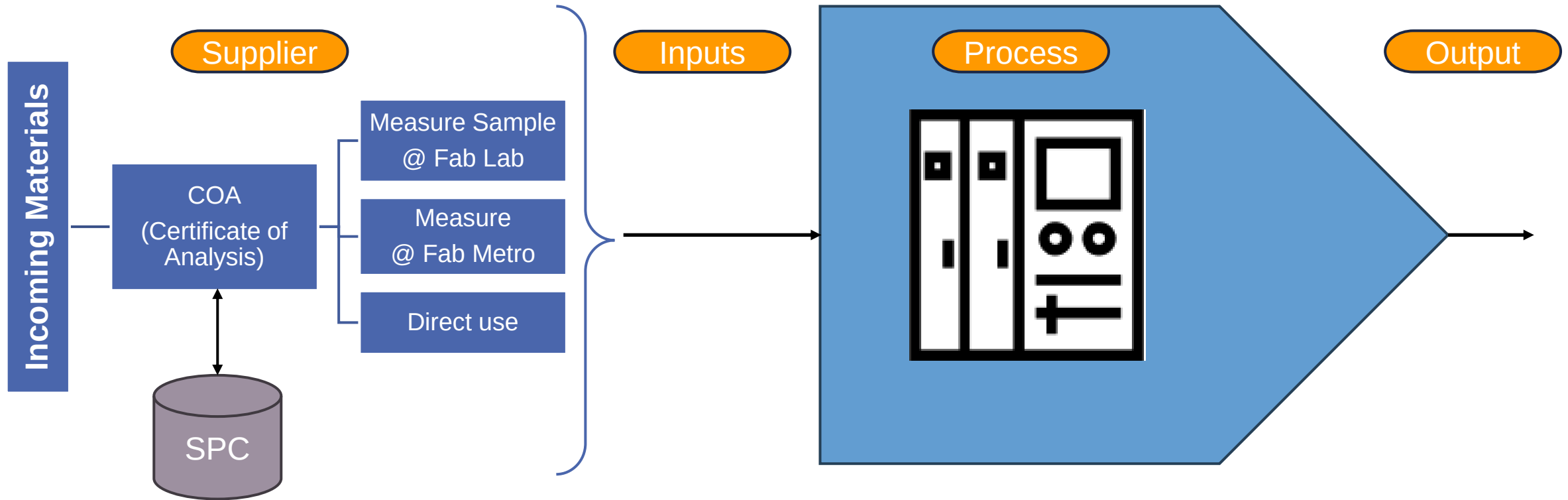


Sources: <http://www.horiba.com/us/en/scientific/products/particle-characterization/applications/cmp/>



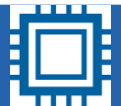
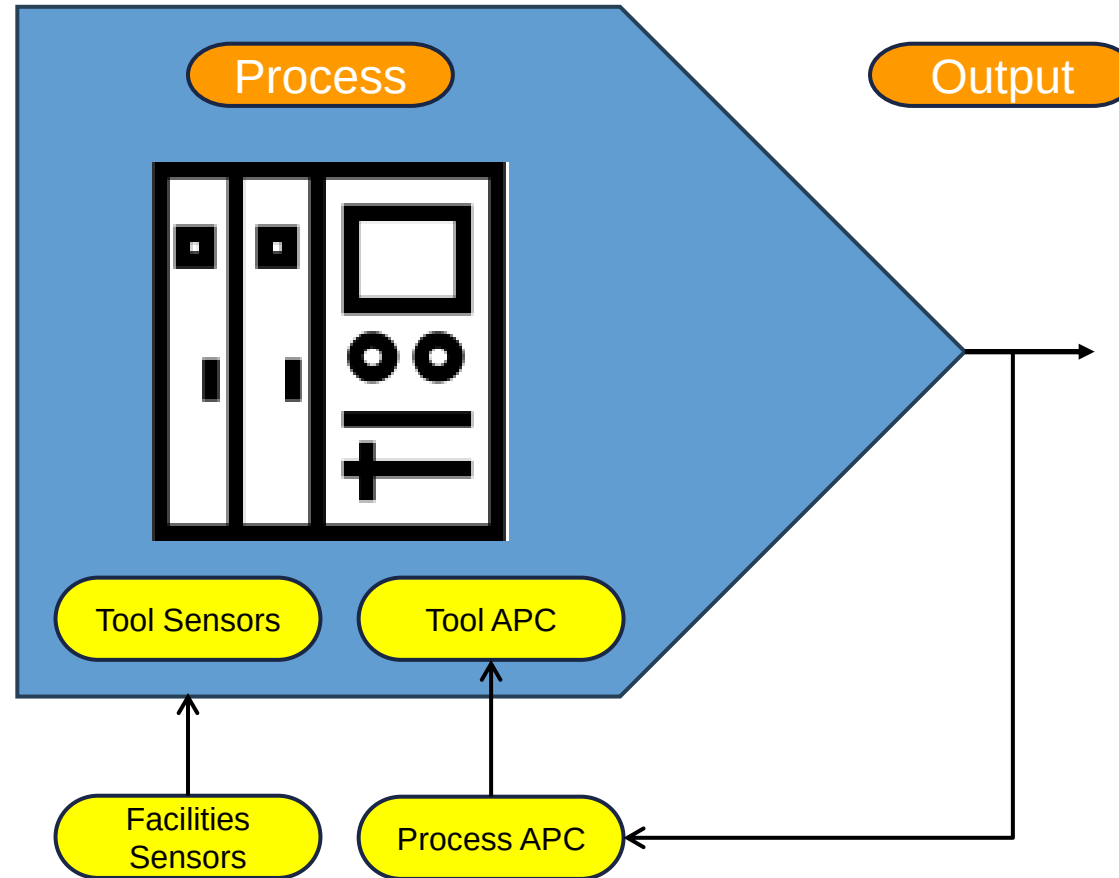
QC_0: Incoming Supplier Q control – Pre-Process

Before the process starts, we must ensure that input quality is passing the requirements for our process



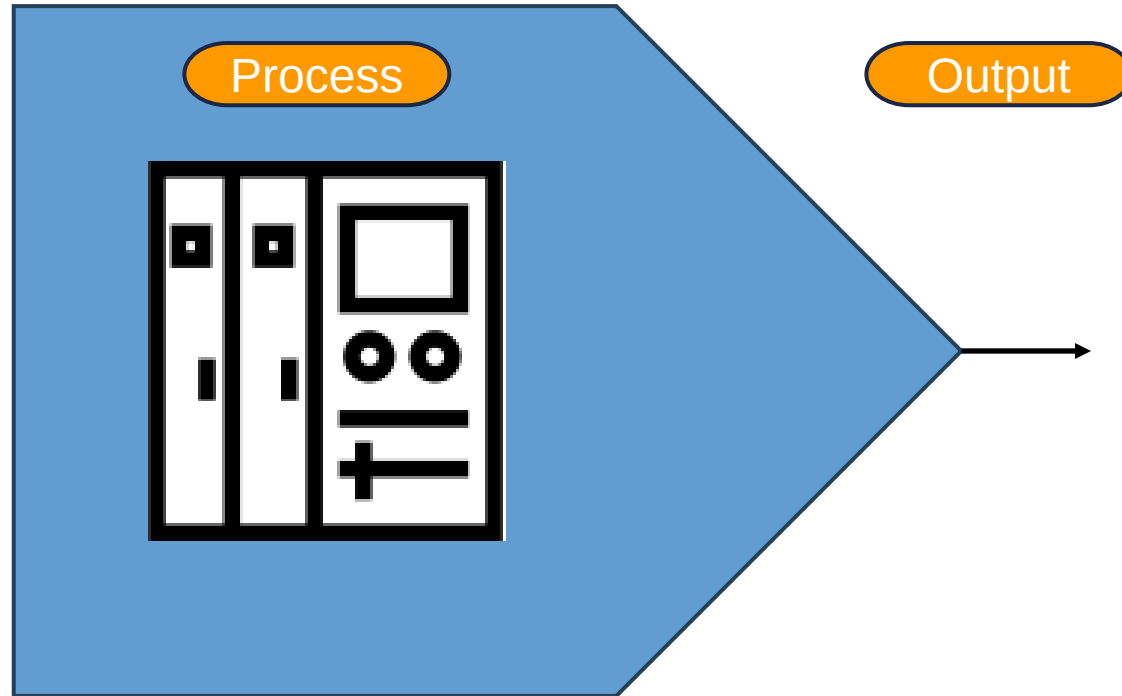
QC_1: Tool Level (Facilities, Tool FDC, APC, etc) - In-Process

During the process, we must ensure we have adequate sensors & feedback loops to produce stable & robust output

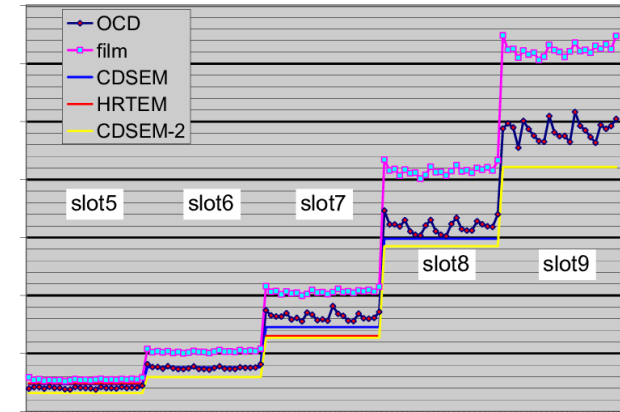


QC_2: Inline Metrology & Defect Screening - Post-Process

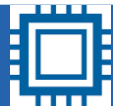
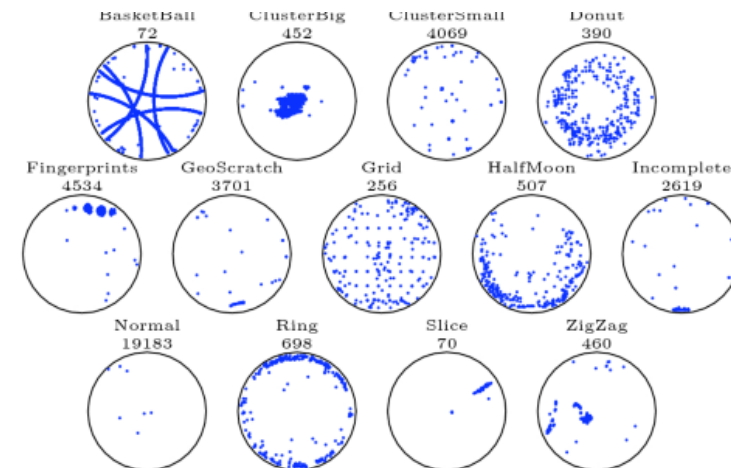
After the process, we need to check whether the output meets the quality requirements, in our case it is process parameters & defectivity



Inline Process Data

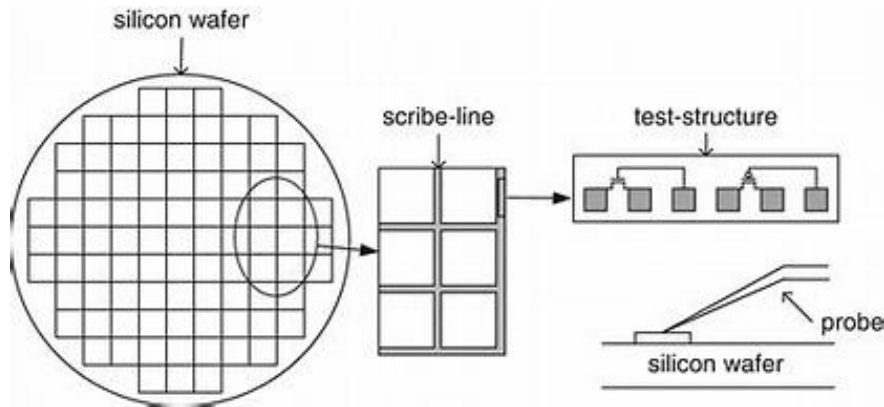


Inline Defect Data



QC_3: Electrical Test (ET) & Reliability Tests

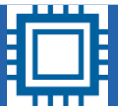
Electrical tests verify the functionality of devices & interconnects. A wafer reliability test is a process used to validate the device's life over time. It involves exposing the wafer to various stress conditions like temperature cycling, humidity, and electrical stress to see how it holds up.



Electrical devices are placed in between the product dies and tested for various critical electrical parameters vs specification limits

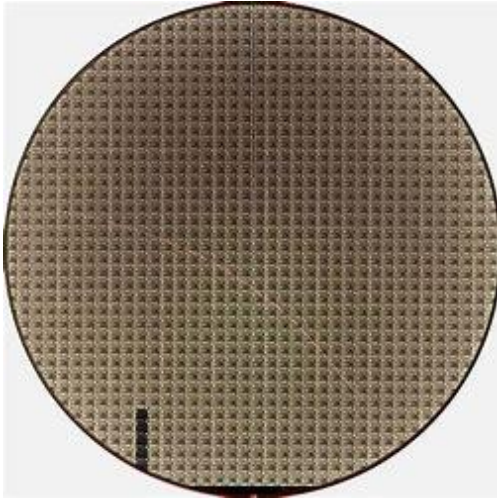


Reliability check is done to inspect that wafer can work reliably without failing in the end-product.

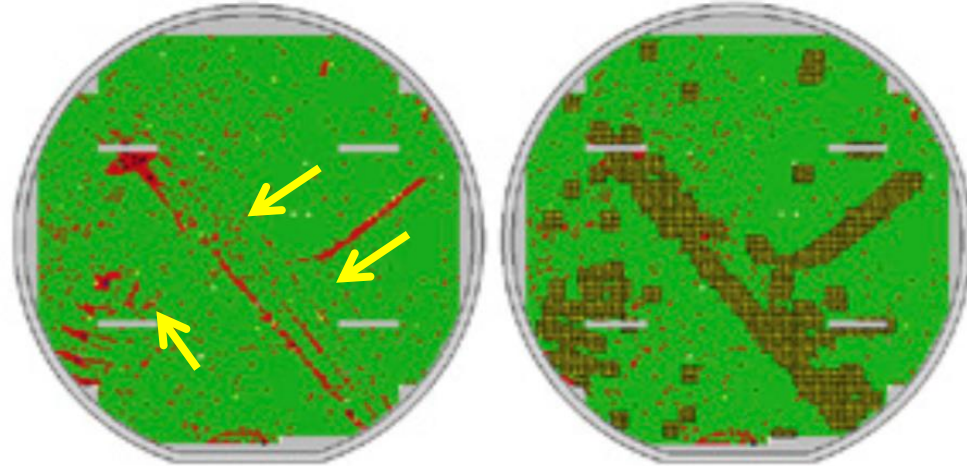


QC_4: Outgoing Quality Control (OQC)

OQC is the process of ensuring that our wafers meet all quality standards before they are shipped to customers or OSATs

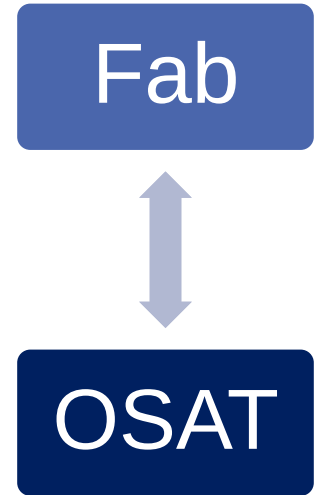


Optical OQC: This refers to quality checks based on visual inspection or optical systems

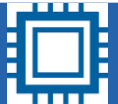


Inking OQC: Inking refers to special marking (nowadays its digital) to identify dies which are problematic and should not be tested to save on test time & costs.

Fab can request Inking based on scratches and reliability risks



B2B OQC: A Wafer Disposition System (WDS) connects Fab with OSATs for communication & traceability of shipped wafers, ink maps and disposition of wafers



QC_5: Wafer Sort Test / Package or Final Test

Wafer sort test identifies good vs non-functional dies, and only good dies are sent forward for packaging. Package test includes tests for package integrity, analog/RF/ESD performance, thermal cycling tests, etc.. Final or System-level test is the last check in chip production on a system board.

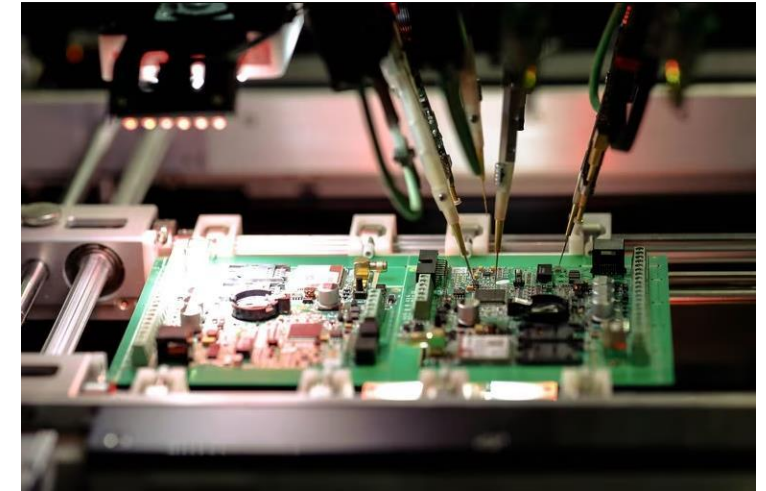
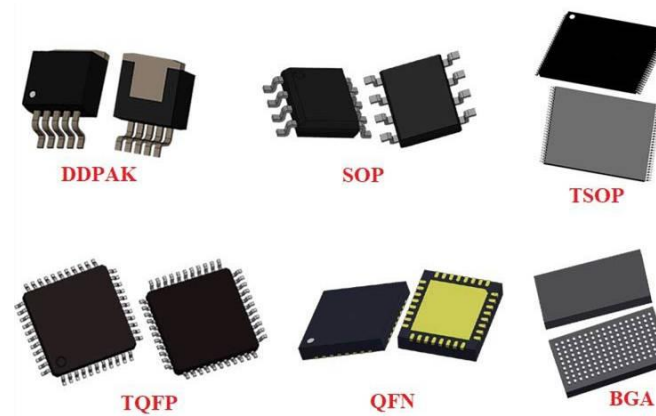
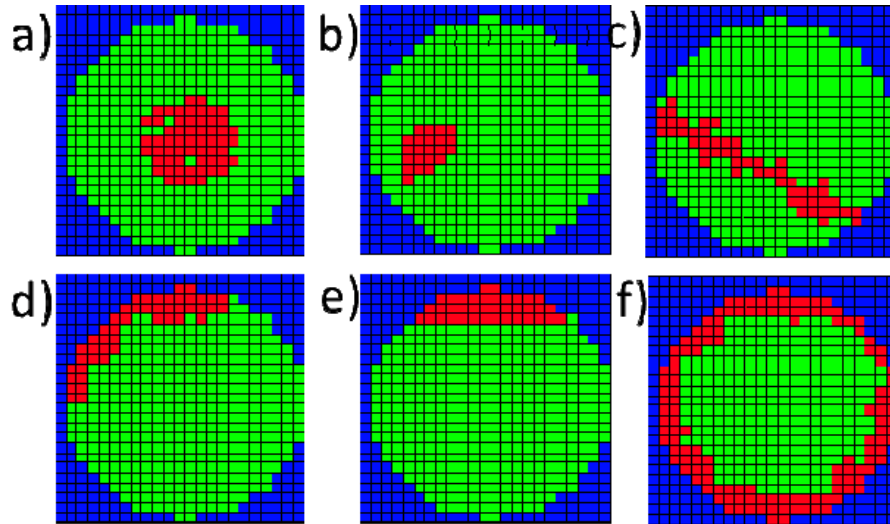


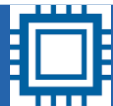
Figure 1. Defect cluster patterns: a) Bulls-Eye, b) Blob, c) Line, d) Edge, e) Hat, f) Ring

Wafer Sort Test & Signatures

Package Test

System Test

Source: <https://mungfali.com/explore/Wafer-Map>



Scrap Costs : Line Yield, Sort Yield, Final Yield

Every step of the chip making process adds value & costs. Detecting defects earlier can save us a lot of money & pain!

Early detection saves money, time & reputation !

4 fully processed wafers =



Fab Line Yield (LY)

Assume we started 1000 wafers & after full process, at the end for shipment, we have 900 good wafers

$$LY = 900/1000 = 90\%$$

Wafer Sort Yield

The shipped 900 good wafers get sort tested for verifying good dies.

$$\text{Wafer Sort Yield} = \frac{\text{Passing dies per wafer}}{\text{Total dies on wafer}}$$

Package Yield +

The good sorted dies from wafer sort test gets packaged, but some good dies may fail at Package test !
While some may fail in the final system/product (e.g. Laptop) !



Quality Policy and the 3-legged stool

Quality Policy defines the organization's commitment to Quality. It needs to encompass customer, employees, product/service commitment and quality framework. Quality is like a 3-legged stool of Supplier, Manufacturing & Customer quality held together by QMS, Labs & Reliability



Vision:

Delivering world-class, reliable, and sustainable electronic solutions through uncompromising quality.

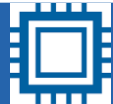
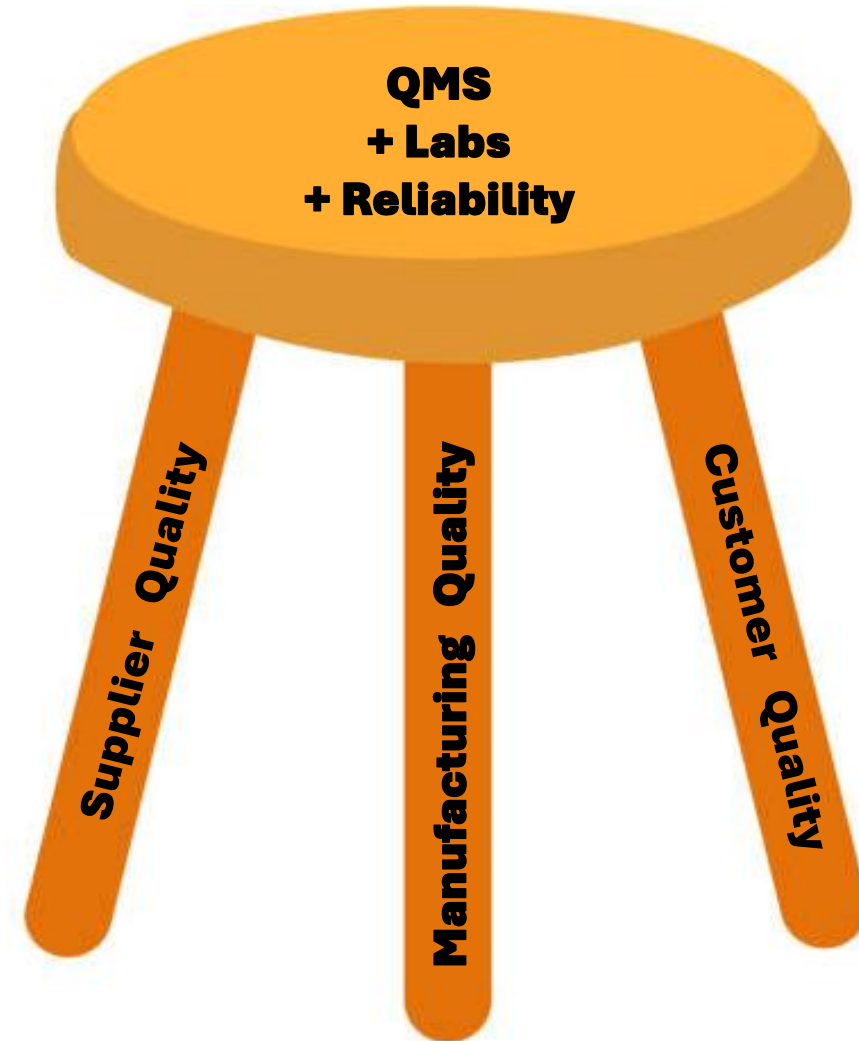
Mission:

Driving innovation, operational excellence, and continual improvement to ensure trust, reliability, and value for our stakeholders.

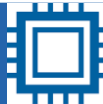
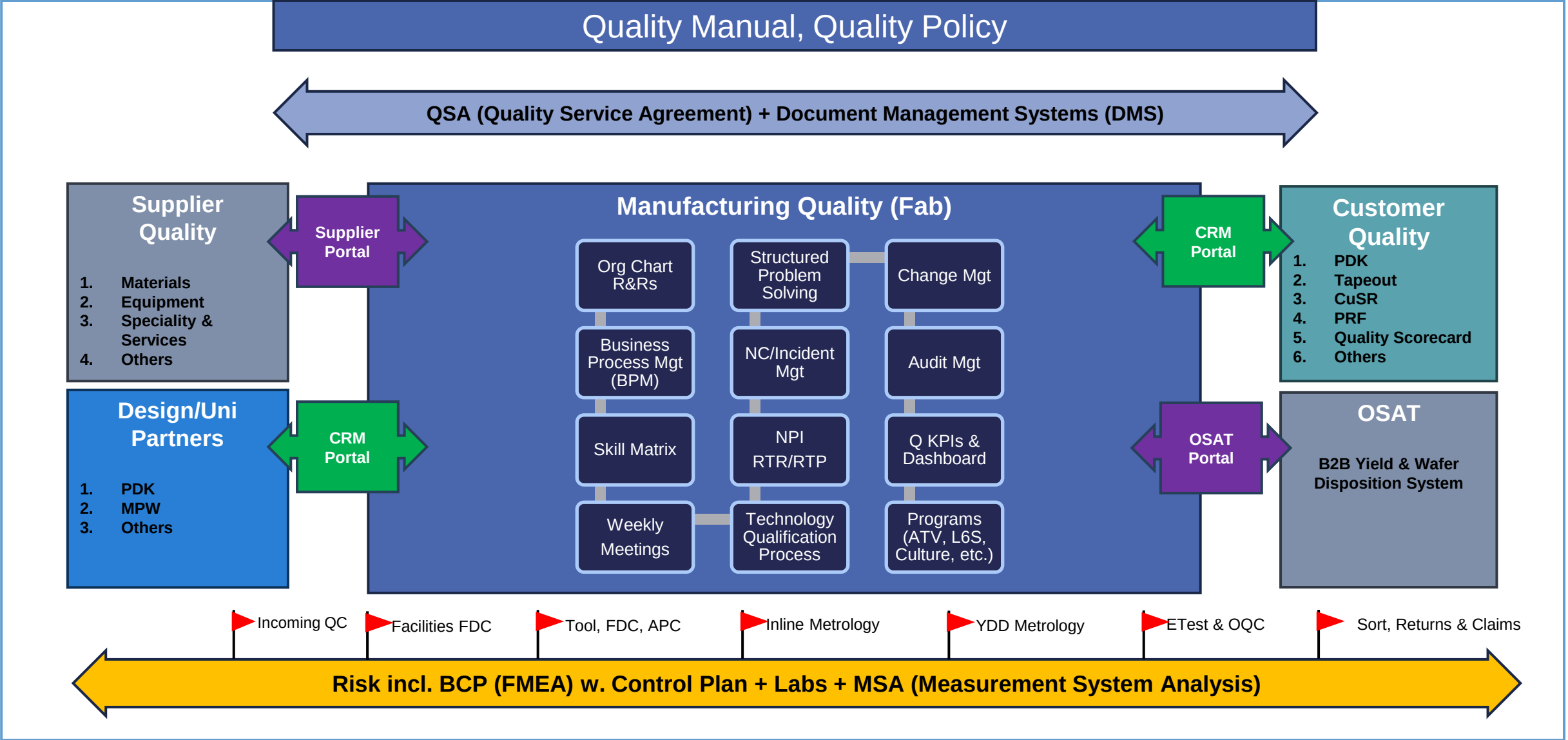
Quality Policy:

We empower our talented workforce to deliver robust, differentiated electronic solutions through flawless execution and continual improvement, ensuring excellence for our customers.

Proposal for TSMPL Q Policy



Semiconductor Foundry QMS Landscape



Quality: Methods & Workflows

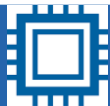
Methods are universal and provide framework for various topics (e.g. Change Management is applicable to all departments). Workflows are set of process steps which deliver an unique service or product output (e.g. Supplier Qualification Workflow)

Methods

1. Event/Incident Management
2. Structured Problem Solving (CA/PA, 8D)
3. Change Management
4. Risk & Control Management
5. Document Management
6. Audit Management
7. Business Process Management (BPM)
8. Claims & Returns Management

Workflows

1. Supplier Qualification
2. Material Qualification
3. Tool/Equipment Qualification
4. Technology Qualification
5. Design IP Qualification
6. Reliability Qualification
7. many more..



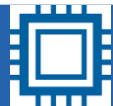
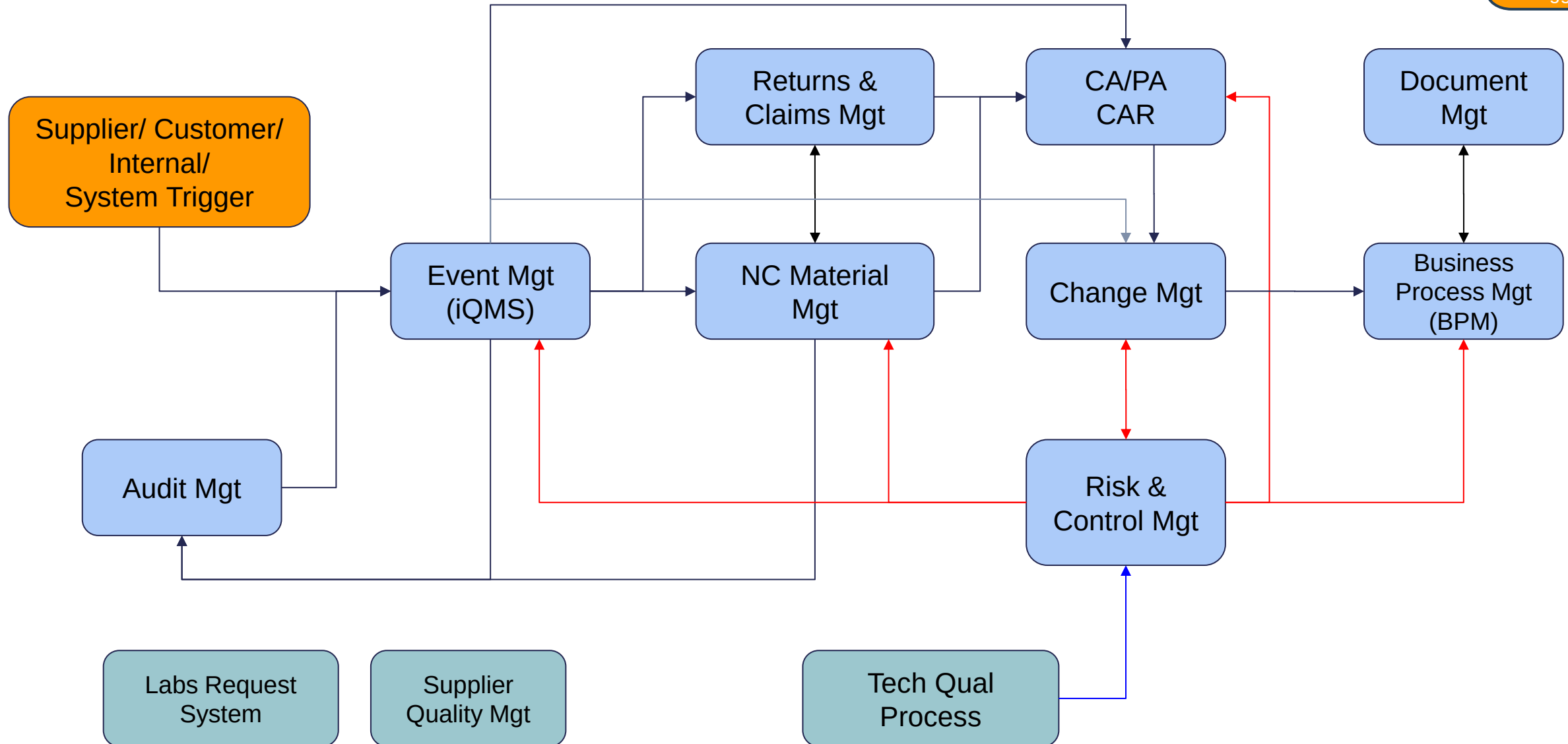
Q Methods Interaction

Q Methods interact with each other, hence the data, applications and systems need to be integrated

Q Method

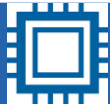
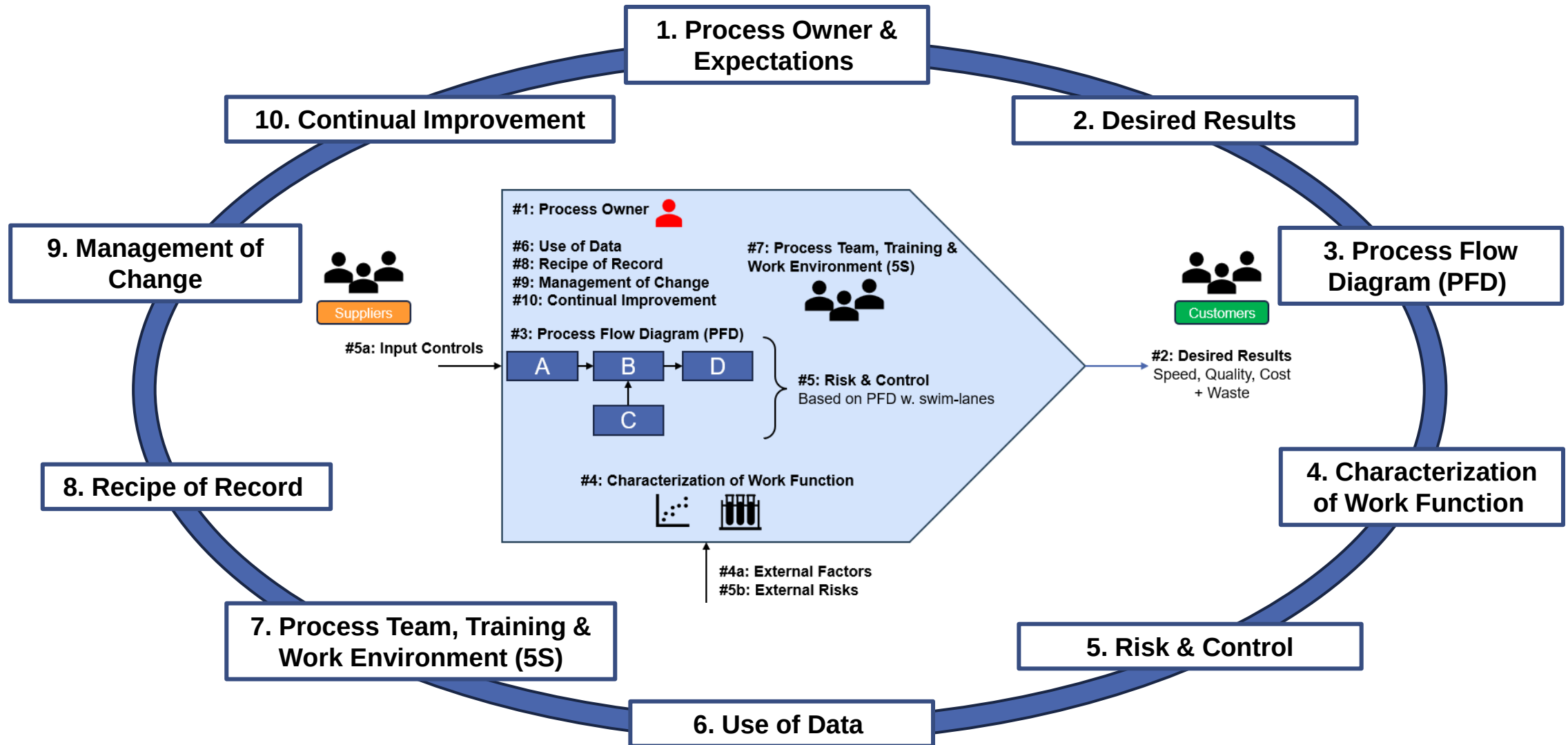
Workflows

Triggers



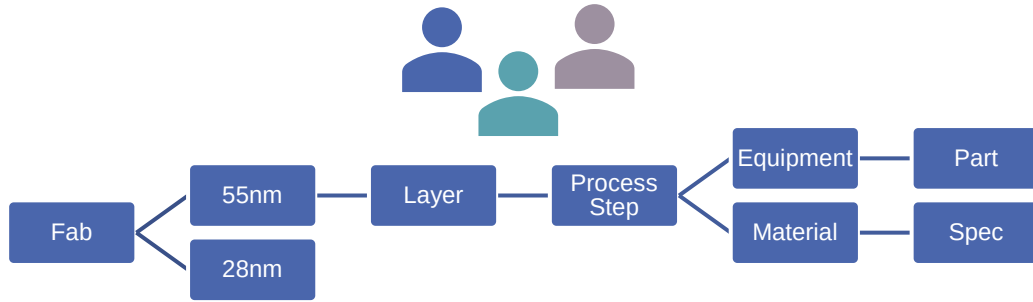
What is Voice of Process (VOP) ?

It is quality framework covering all critical aspects to build a robust & excellent processes

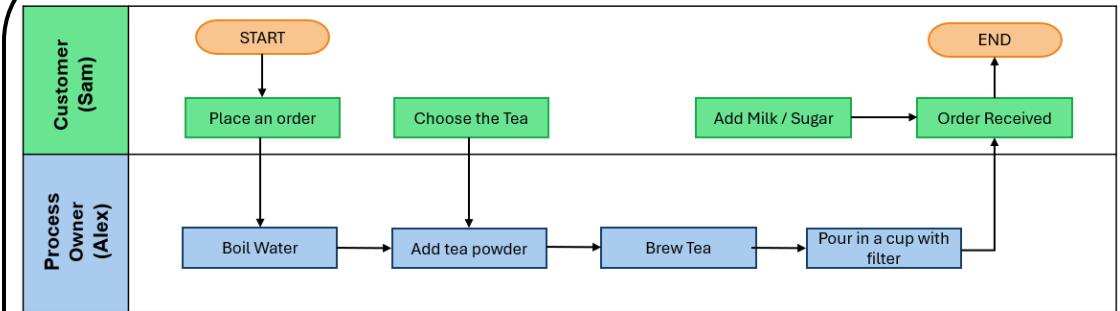


Quality: 4 Tips on building Lean Processes!

Entities & Object Design + Process Flow Diagrams w. Swim lanes + Value Add vs No-Value Add + Lifecycle Approach



Minimize the number of Entities (Humans, systems) interacting with the process and use Object-based data model



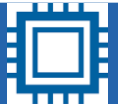
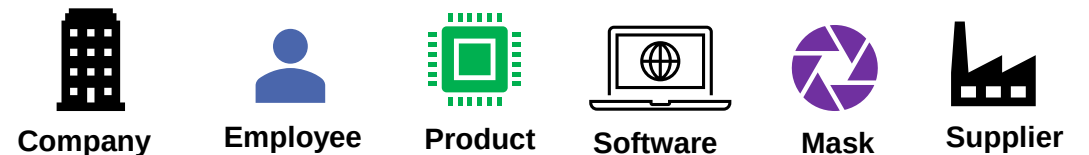
Build the Process Flow Diagram and then add the Entities which interact with your process as Swim-lanes

Imagine a employee vacation approval process as below..
Which step is Value-Add vs Non-Value Add ?



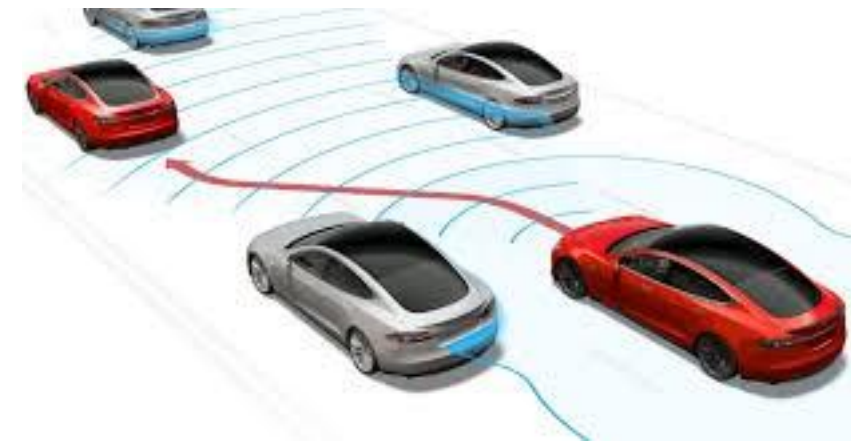
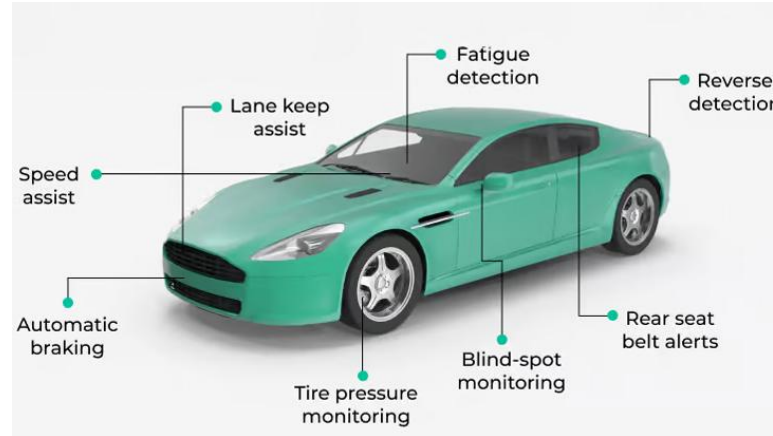
Analyse each process step whether it adds functional value or not.
Remove non-value add steps!

For every "Object" use Lifecycle approach:
Initiation → Qualification → Operation → EOL



Process Maturity Continuum

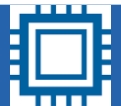
Grading of process maturity and robustness is based on the continuum of correcting, controlling, improving, preventing & anticipating (CCIPA). We must build compliant & best-in-class robust processes!



Compliant /
Correcting

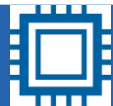
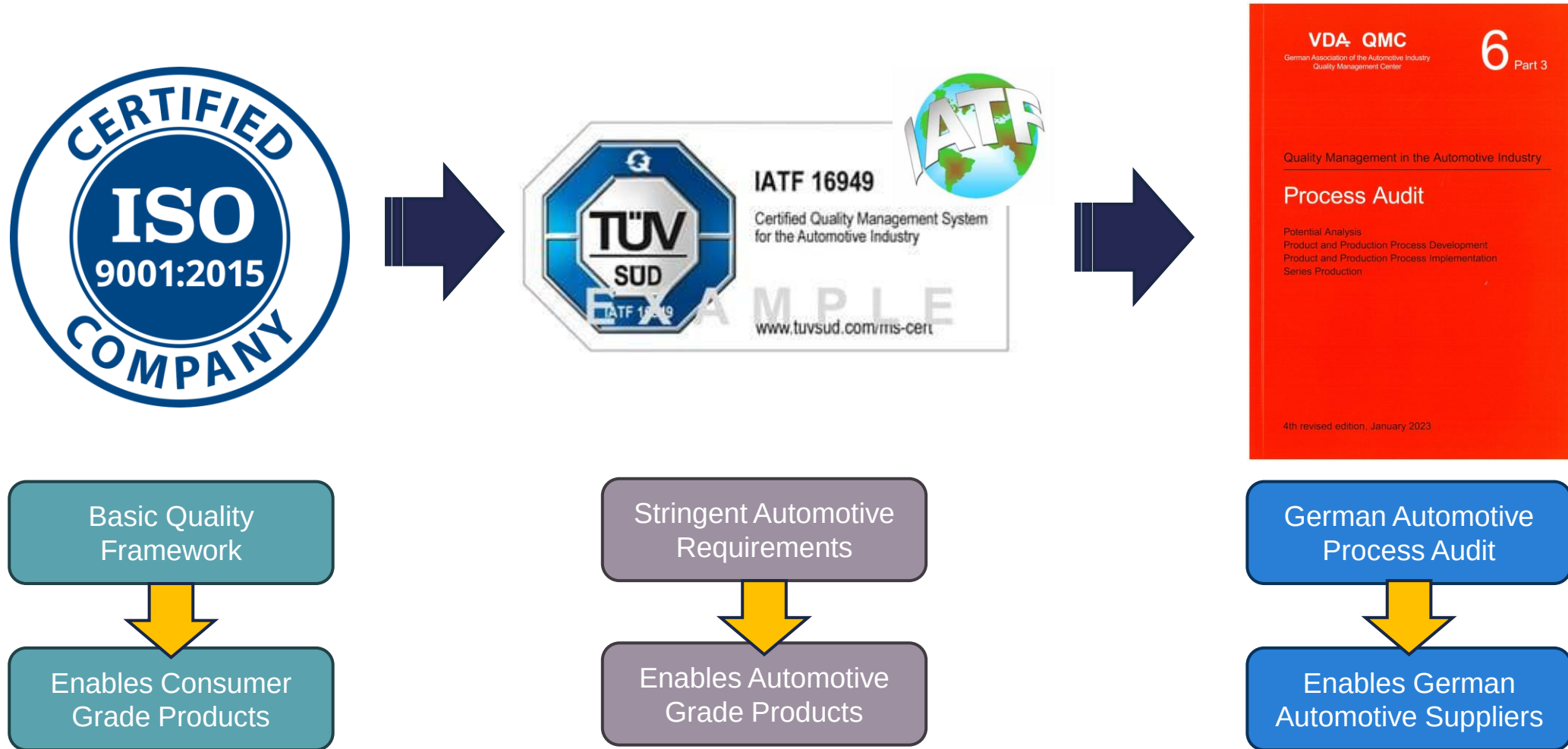
Controlling /
Improving /
Preventing

Anticipating /
Predicting

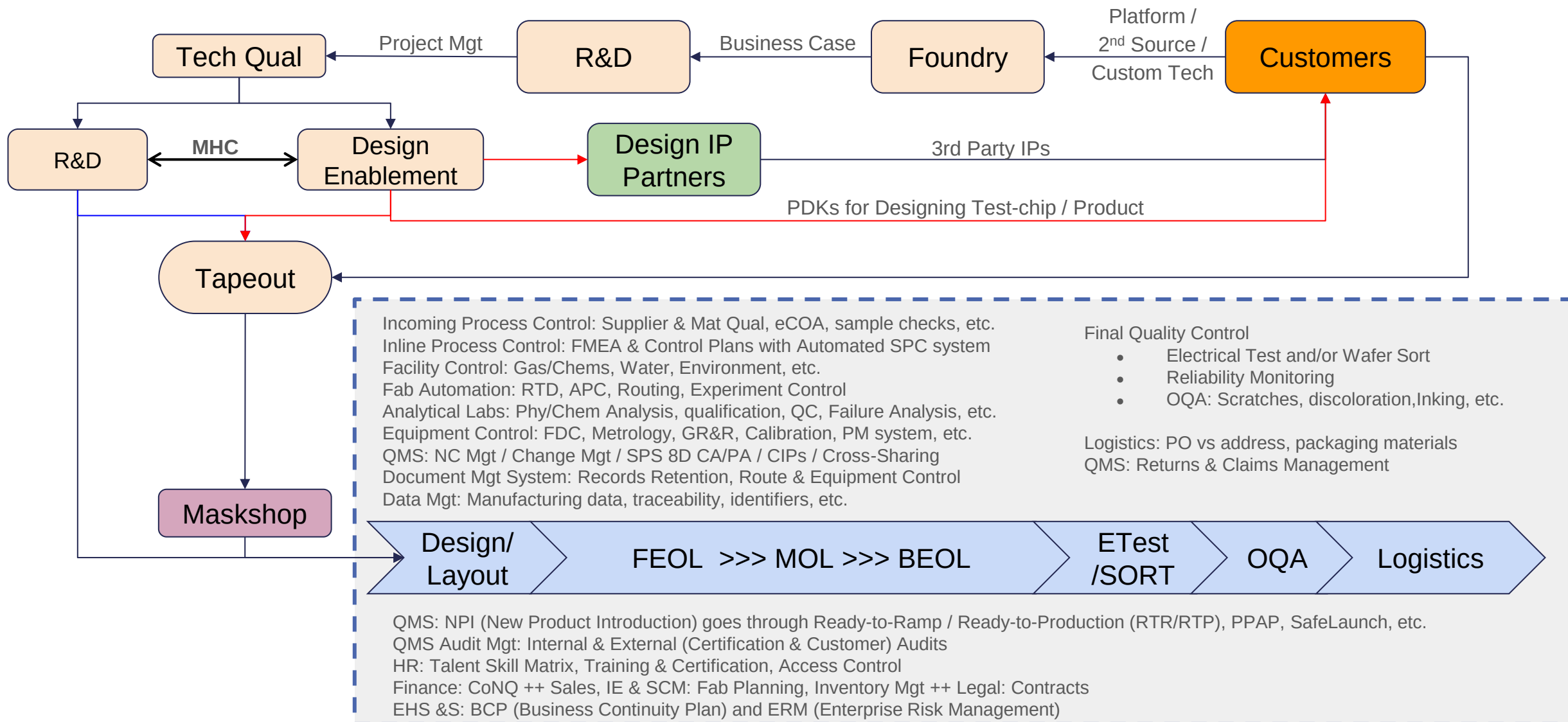


Audit, Compliance & Certification

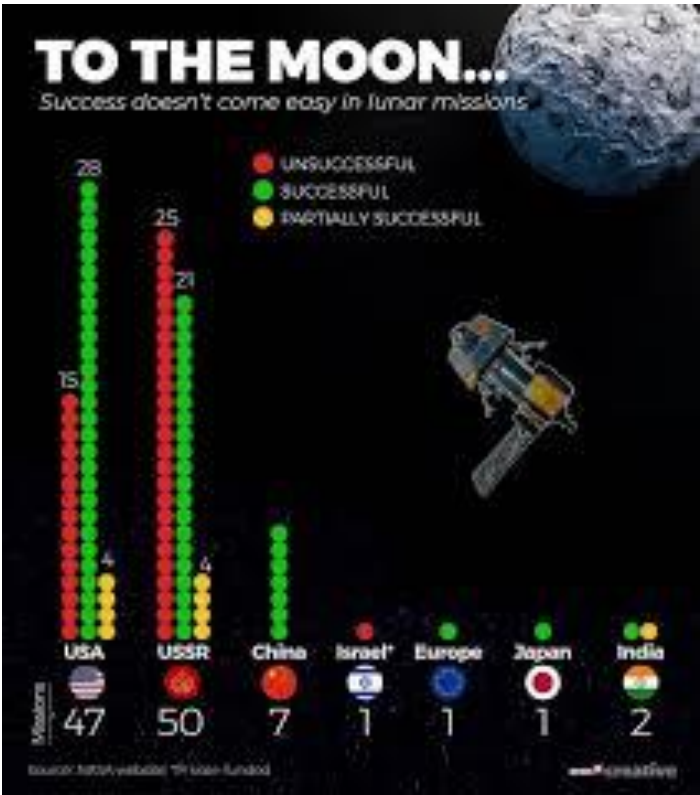
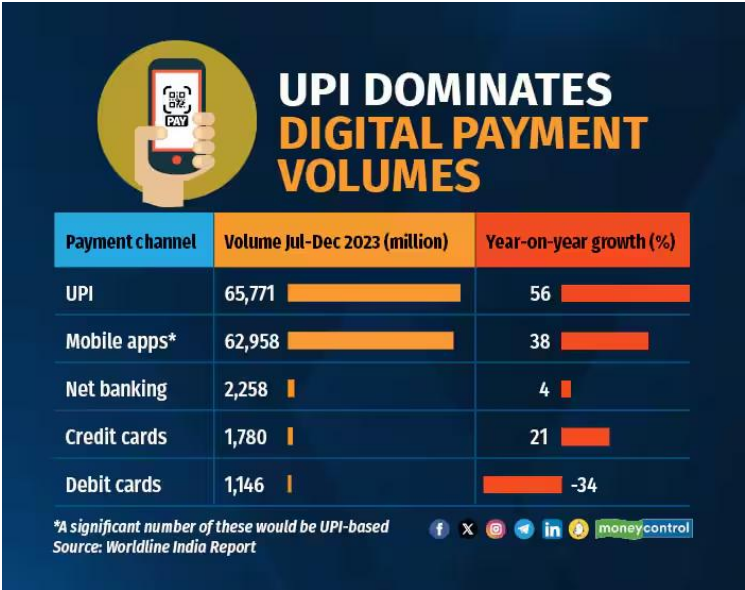
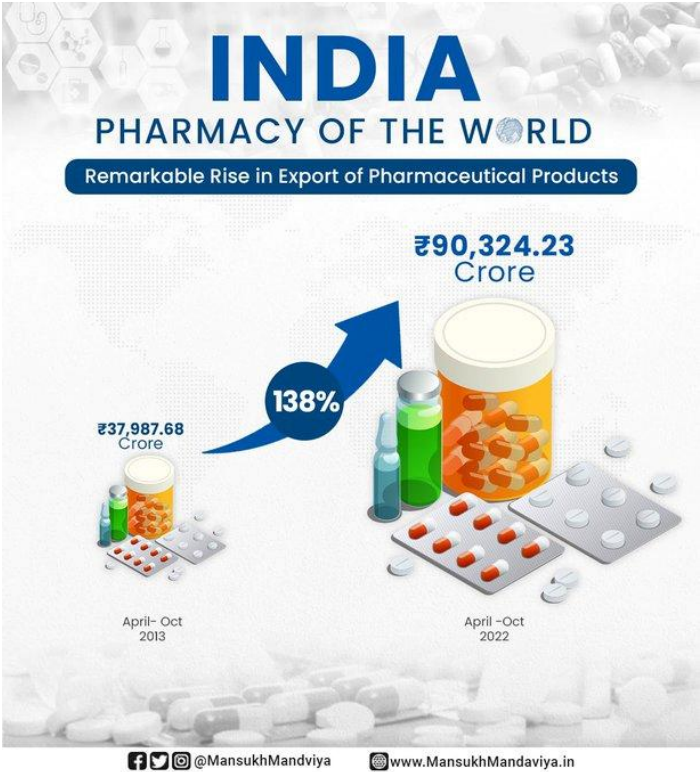
Semiconductor businesses need to ensure that the organization, products/services are compliant to stringent industry compliance certifications like ISO9001, IATF-16949 incl. AIAG Core Tools (Automotive), VDA 6.3 (German Automotive) and other niche standards (e.g. AS9001 for Aerospace)



Bringing it together: High-level E2E Semiconductor Foundry Workflow



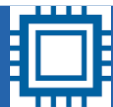
But there is hope... Excellence “Made in India”



Acronyms used in this deck 😊

AEC	Automotive Electronics Council	ISO	International Organization for Standardization
AIAG	Automotive Industry Action Group	JEDEC	Joint Electron Device Engineering Council
ALD	Atomic Level Deposition	LP	Low Power
APC	Advanced Process Control	LPP	Low Power Plus
ASIC	Application Specific Integrated Circuit	LVS	Layout Versus Schematic
B2B	Business to Business	LY	Line Yield
BCP	Business Continuity Plan	MHC	Model Hardware Correlation
BEOL	Back End of Line	MCU	MicroControllerUnit
BPM	Business Process Management	MPW	Multi Project Wafer
CA	Contact	MSA	Measurement System Analysis
COA	Certificate of Analysis	nm	Nanometer
Cp/Cpk	Process Capability Index	NPI	New Product Introduction
CRM	Customer Relationship Management	NVM	Non-Volatile Memory
Cu	Copper	NW	N-Well
CuSR	Customer Specific Request	OPC	Optical Proximity Correction
CVD	Chemical Vapour Deposition	OSAT	Outsourced Semiconductor Assembly and Test
D0	Defect Density	PC	Poly (Gate) mask
DFM	Design For Manufacturing	PCB	Printed Circuit Board
DI	De-Ionized	PDK	Process Design Kit
DM	Design Manual	PEX	Parametric Extraction
DMS	Document Management System	PR	PhotoResist
DRC	Design Rule Check	PRF	Process Request Form
ECD	Electro Chemical Deposition	PVD	Physical Vapour Deposition
EM	Electromigration	PVS	Physical Verification System
EOL	End of Life	QC	Quality Control
ERA	Early Reliability Assessment	QMS	Quality Management System
FDC	Fault Detection & Classification	QSA	Quality Service Agreement
FDSOI	Fully-Depleted Silicon On Insulator	RTL	Register Transfer Level
FEOL	Front End of Line	SADP	Self-Aligned Double Patterning
FET	Field Effective Transistors	SG	Stage Gate
FMEA	Failure Modes and Effects Analysis	SPC	Statistical Process Control
GDS	Graphic Design Sytem	SPICE	Simulation Program with Integrated Circuit Emphasis
GTC	GPU Technology Conference	SRAM	Static Random Access Memory
H/M C	High/Medium Current	STA	Static Timing Analysis
HE	High Energy	TSMPL	TATA Semiconductor Manufacturing Private Limited
HOD	Head of Department	VOP	Voice of Process
IATF	International Automotive Task Force	YDD	Yield Defect Density
IC	Integrated Circuit		
IP	Intellectual Property		
IR	InfraRed		

[Acronyms List.xlsx](#)





Thank You!

